

Mean-Field Game Strategies for Optimal Execution*
previously titled: **Mean-Field Game Strategies for a Major-Minor Agent Optimal Execution Problem**

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Abstract. Algorithmic trading strategies for execution often focus on the individual agent who is liquidating/acquiring shares. When generalized to multiple agents, the resulting stochastic game is notoriously difficult to solve in closed-form. Here, we circumvent the difficulties by investigating a mean-field game framework containing (i) a major agent who is liquidating a large number of shares, (ii) a number of minor agents (high-frequency traders (HFTs)) who detect and trade against the liquidator, and (iii) noise traders who buy and sell for exogenous reasons. Our setup accounts for permanent price impact stemming from all trader types inducing an interaction between major and minor agents. Both optimizing agents trade against noise traders as well as one another. This stochastic dynamic game contains couplings in the price and trade dynamics, and we use a mean-field game approach to solve the problem. We obtain a set of decentralized feedback trading strategies for the major and minor agents, and express the solution explicitly in terms of a deterministic fixed point problem. For a finite N population of HFTs, the set of major-minor agent mean-field game strategies is shown to have an ϵ_N -Nash equilibrium property where $\epsilon_N \rightarrow 0$ as $N \rightarrow \infty$.

Key word. Algorithmic trading, high-frequency trading, optimal execution, stochastic optimal control, mean-field games and market microstructure theory.

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1. Introduction. Algorithmic trading refers to any form of quantitative trading strategy which is executed by computer programs that make decisions automatically. An important class of algorithmic trading strategies is high-frequency trading which are algorithms that are executed at a very high speed. Automated trading algorithms have grown rapidly and are now applied across electronic trading platforms worldwide. As a result, there has been a considerable amount of research devoted to the modeling and design of efficient trading strategies. In particular, two related but distinct lines of research include optimal execution problems and optimal limit order placements. In optimal execution problems, an agent is required to choose an efficient trading strategy for liquidating/acquiring a portfolio containing a large quantity of a given security in order to maximize her expected wealth (by minimizing cost or maximizing profit). As such, the agent aims to balance the tradeoff between trading slowly to reduce market impact, and execution cost, and trading fast to reduce the risk of future uncertainty in prices (see e.g., [1, 3, 20, 27, 28, 34] among others). Optimal limit order placement aims instead to benefit from buying low and selling high and profits from the bid-ask spread without being adversely selected. The aim is to optimally place bid and ask limit orders to maximize the expected profit and here the agent faces a tradeoff between trading gains and non-execution risks (see e.g., [4, 5, 17, 18, 19, 22, 23, 29, 30], among others).

Most of the literature on algorithmic trading focuses on the single agent setting. Part of the reason for this focus is that accounting for multiple agents leads to high dimensional problems and complicated interactions, and typically renders the analysis intractable using standard techniques in stochastic optimal control. This paper develops the first stochastic game framework, and solves for the optimal strategies, between optimizing agents in electronic exchanges¹.

There has, however, been some recent work in addressing multiple agent problems, see e.g., [7, 15, 37, 43, 10], among others. In particular, [15] develop a closed-loop Nash equilibrium strategy for a model consisting of a distressed agent who needs to liquidate shares and is facing a predator trader with longer time constraints. A similar setting to [15] is investigated in [37] who study an agent who needs to liquidate a large number of shares and is facing an uninformed arbitrageur who hopes to profit from the liquidator's trading by learning from observed price fluctuations. [43] develop an open-loop Nash equilibrium strategy for a finite population of liquidators who are coupled through fundamental price dynamics and aim to maximize their individual mean-variance utility functions using continuous trading.

Large population dynamic multi-agent competitive phenomena occur in various other fields including communication networks, biological systems, economics and social sciences. In particular, an important class of large-scale dynamical systems is that of dynamic games with a large number of minor agents in which each agent interacts with the average (or so-called mean-field) effect of other agents via couplings in their individual dynamics or individual cost functions. A minor agent is an agent which, asymptotically as the population size goes to infinity, has a negligible influence on the overall system while the overall population's effect on it is significant. For large population stochastic dynamic games with mean-field couplings and no major agent, the so-called mean-field game (MFG) theory was originally developed as a decentralized methodology in a series of papers by Huang together with Caines and

¹The first version of this paper was already available online in 2015.

Malhamé (see [32, 33], among others). For this class of game problems a closely related approach has been independently developed by [36]. [11] develop a probabilistic approach to MFG problems using tailor-made forms of the stochastic maximum principle. [31, 38, 39] address MFG interactions in the presence of a major agent who has a strong influence on the minor agents. As in these settings, all minor agents respond to the same major agent, their mean-field is directly impacted by the major agent and hence is a random process even asymptotically as the population size of minor agents approaches infinity. This is in contrast to the situation in the standard MFG models with only minor agents. [13] provide an analysis of MFG problems with common noise and demonstrate existence and uniqueness. The reader is referred to the recent survey paper [8], excellent two-volume [12] book, and brief [6] for further developments.

The key idea of the MFG theory is to specify a certain equilibrium relationship between individual strategies and the overall effect of the population on a given agent, called the mass effect, as the population size N goes to infinity. Specifically, in equilibrium: (i) the individual strategy of each agent is a best response to the infinite population mass effect in the sense of a so-called ϵ -Nash equilibrium, and (ii) the set of strategies collectively replicates the same mass effect, this being a dynamic game theoretic fixed point property. The defining ϵ -Nash equilibrium property of the MFG equilibrium indicates that any given agent can benefit at most by ϵ when deviating unilaterally from its MFG strategy. The estimates in [33, 32] show that $\epsilon = O(N^{-\frac{1}{2}})$.

This paper is the first to solve a MFG modeling framework for a multiple agent optimal execution problem with continuous trading. We generalize the classical single agent optimal liquidation problem (as in [3]) to a setting with (i) a major agent who is liquidating a large portion of shares, (ii) a number of minor agents (high-frequency traders (HFTs)) who detect and trade (optimally) along with the liquidator, and (iii) noise traders whose strategies are independent of the major and minor agents' trading strategies. These agents are similar to those in [35], with the major and minor agents acting as informed traders. In this paper, we present a continuous-time formulation of the model, where the major and minor agents can distinguish between the order-flow coming from noise traders, and order-flow coming from the major/minor agents.

In this formulation all agents' (major, minor and noise) trades induce a permanent price impact on the fundamental asset price, and major/minor agents receive a temporary price impact when trading. Each agent aims to balance their price impact against price uncertainty and maximize profits all the while being subject to the action of other traders. This cross-interaction between the major and minor agents (and among the minor agents), through the impact each trader has on the fundamental price's drift, is a distinct feature of our setup.

Our formulation, therefore, falls within the realm of stochastic dynamic game problems with mean-field couplings in the dynamics. It is not feasible for each individual agent to track the actions of all other agents. To address this issue, we instead analyze the problem using MFG theory and obtain a set of decentralized feedback trading strategies for major and minor agents. The strategies are obtained from a set of auxiliary stochastic optimal control problems which we explicitly solve. The MFG solution is then reduced to a deterministic fixed point problem – which we do not solve explicitly, but there are efficient numerical

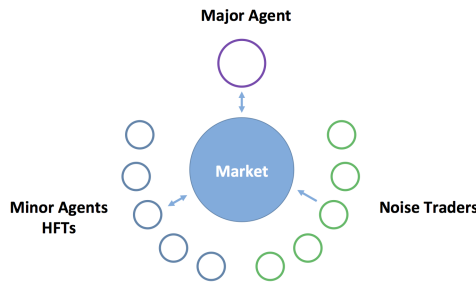


Figure 1. Interaction between major and minor agents through the market. The major and minor agents optimize and account for the actions of the noise traders as well as market information, while the noise traders are not optimizing agents.

schemes for it. It is important to note that the asymptotic mean-field of HFTs is a “controlled mean-field” depending on the major agent’s trading strategy which is in contrast to both non-controlled deterministic mean-field of standard MFG problems in [33, 32]. The resulting mean-field control laws yield a trading algorithm so that each individual agent may implement a decentralized strategy based on their local information together with information on the mean of minor agents’ initial shares. We also show that for a finite N population of HFTs, the set of major-minor agent MFG strategies has an ϵ_N -Nash equilibrium property where $\epsilon_N \rightarrow 0$ as $N \rightarrow \infty$.

Recently, [14] provide as an example a MFG formulation of an execution problem, however, do not analyze the solution, nor does it contain noise traders or major agents. While [9] investigate a MFG formulation for a sea of optimizing minor agents (as each agent contributes N^{-1} to price impact) without the major agent or sea of noise traders. [24, 25] look a MFG formulation (without a major agent) for algorithmic trading with latent factors using a convex analysis approach.

The remainder of this paper is organized as follows. Section 2 is dedicated to the formulation of the major-minor agent optimal execution problem. An MFG approach to the problem is presented in Section 3, which includes two auxiliary stochastic optimal control problems for major and minor agents in the mean-field limit and their solutions (Subsection 3.1 and 3.2), a MFG consistency condition (Subsection 4.1). Section 5 shows the ϵ -Nash equilibrium property of MFG decentralized trading strategies. Sample numerical simulations are given in Section 6, and some concluding remarks are provided in Section 7.

2. Major–Minor Agent Optimal Execution Problem. We consider a dynamic game system involving: (i) a major institutional investor agent, (ii) a large population of minor agents as informed high-frequency traders, and (iii) a large population of noise traders. A diagram of the interaction between major and minor agents through the market is shown in Figure 1. Here, we show noise traders who buy and sell at “random” and add in additional order-flow. The description of each part of the model is given in detail below.

Throughout, we use the following notation. The trading horizon is $T \in (0, \infty)$ and $\mathfrak{T} := [0, T]$. Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in \mathfrak{T}}, \mathbb{P})$ be a complete filtered probability space on which we have a standard $\mathbb{P}$ -martingale $M = (M_t)_{t \in \mathfrak{T}}$ and a noise trader order-flow process $\mu = (\mu_t)_{t \in \mathfrak{T}}$ (assumed independent of M), and where $\mathcal{F}_t$ is the natural Filtration generated by M and μ . $\mathbb{E}$ denotes the expectation operator. The set of non-negative real numbers is denoted by $\mathbb{R}_+$. The integer N denotes the population size. The superscript N for a process (such as state, control or wealth process) indicates the dependence on the population size N . We use the

superscript 0 for the major agent and an integer valued superscript for an individual minor agent. $\mathcal{C}_b(S)$ is the set of bounded and continuous functions on S . For two functions f, g defined on some subset of $\mathbb{R}$ we use the notation $f(x) = O(g(x))$ if and only if there exist constants C and M such that $|f(x)| \leq C|g(x)|$ for all $x > M$.

2.1. Major and Minor Agents Trading Dynamics. We denote the major agent's inventory by q , the minor agent- i 's inventory by q_i , the average inventory of minor agents by y , and the trading speed of noise traders by u .

2.1.1. Major Agent's State and Control Processes. By the end of the trading horizon T , the major agent aims to liquidate her inventories $Q_0^{\nu_0} = (Q_{0,t}^{\nu_0})_{t \in \mathfrak{T}}$. She does so by controlling her continuous rate of trading $\nu_0 = (\nu_{0,t})_{t \in \mathfrak{T}}$ so that

$$(2.1) \quad dQ_{0,t}^{\nu_0} = \nu_{0,t} dt, \quad t \in \mathfrak{T}.$$

$F^\nu = (F_t^\nu)_{t \in \mathfrak{T}}$ denotes the fundamental asset price process (i.e. midprice of the asset, and present the specific model in Section 2.3). The notation F^ν denotes the dependence of this fundamental price on the trading rates of all agents: $\nu = (\nu_0, (\nu_i)_{i \in \mathfrak{N}})$ where $i \in \mathfrak{N} := \{1, \dots, N\}$. The price at which the major agent sells the asset for, called the execution price, when trading at a rate $\nu_{0,t}$ at time t is

$$(2.2) \quad S_{0,t}^\nu = F_t^\nu + a_0 \nu_{0,t}, \quad t \in \mathfrak{T}.$$

Here, $a_0 > 0$ models the (linear) temporary price impact that the major agent's trading has on the price. The reader is referred to [34] and [20] for an illustration of how the parameter a_0 may be estimated from market data. The major agent's cash process $X_0^{\nu_0} = (X_{0,t}^{\nu_0})_{t \in \mathfrak{T}}$ is

$$(2.3) \quad X_{0,t}^{\nu_0} = x_0 - \int_0^t S_{0,u}^\nu \nu_{0,u} du, \quad t \in \mathfrak{T},$$

with given initial cash $X_{0,0}^{\nu_0} = x_0$ – typically set to 0.

2.1.2. Minor Agents' State and Control Processes. Next, we consider a large number of minor agents, indexed by $i \in \mathfrak{N} := \{1, \dots, N\}$, who are opportunistic HFTs. They trade at a rate $\nu_i = (\nu_{i,t})_{t \in \mathfrak{T}}$ – which may be positive or negative depending on whether the agent is selling or buying inventories, respectively. As such, their controlled inventory process $Q_i^{\nu_i} = (Q_{i,t}^{\nu_i})_{t \in \mathfrak{T}}$ satisfies

$$(2.4) \quad dQ_{i,t} = \nu_{i,t} dt, \quad i \in \mathfrak{N}, \quad t \in \mathfrak{T}.$$

The set of initial shares of HFTs is $\{Q_{i,0}\}_{i \in \mathfrak{N}}$. Similar to the major agent, minor agent- i receives an execution price

$$(2.5) \quad S_{i,t}^{\nu_i} = F_t^\nu + a \nu_{i,t}, \quad i \in \mathfrak{N}, \quad t \in \mathfrak{T},$$

when trading at a rate of $\nu_{i,t}$ at time t , where $a > 0$ models the (linear) temporary impact that the trading has on the midprice of the asset. This impact may differ from, and is likely smaller than, the major agent's temporary impact parameter, as the HFTs may be able to better take advantage of order-book and order-flow information.

Remark 2.1. We assume homogenous minor-agents with the same parameter a in the execution price process (2.5). However, the modeling may be generalized to the case of heterogeneous minor agents with different temporary impact parameters $\{a_i : i \in \mathfrak{N}\}$ as a sequence of independent random variables drawn from a known prior distribution.

Minor-agent- i 's cash process $X_i^{\nu_i} = (X_{i,t}^{\nu_i})_{t \in \mathfrak{T}}$ is

$$(2.6) \quad X_{i,t}^{\nu_i} = x_i - \int_0^t S_{i,u}^{\nu_i} \nu_{i,u} du, \quad i \in \mathfrak{N}, \quad t \in \mathfrak{T},$$

with given initial cash $X_{i,0}^{\nu_i} = x_i$.

While we use a linear price impact model, as in [3] and [40], more general limit order book dynamics have been investigated in [2] and [42].

2.1.3. Agent's Initial Inventory. Before proceeding, we required two assumptions on the minor agents' initial inventories:

- (A1) The initial inventories of minor agents $\{Q_{i,0} : i \in \mathfrak{N}\}$ are independent random variables. Moreover, there exists a constant k independent of N such that $\sup_{i \in \mathfrak{N}} \mathbb{E}|Q_{i,0}|^2 \leq k < \infty$.
- (A2) There exists a distribution function $\mathbf{F}(q)$ on $\mathbb{R}$ such that the sample average of minor agents' initial shares converges almost surely to its expected value, i.e.

$$\lim_{N \rightarrow \infty} \int_{\mathbb{R}} q d\mathbf{F}_N(q) = \int_{\mathbb{R}} q d\mathbf{F}(q) := \mathcal{Q}_0,$$

where $\mathbf{F}_N(q) := (1/N) \sum_{i=1}^N 1_{\{Q_{i,0} < q\}}$ denotes the set of empirical distribution functions.

If the sequence $\{Q_{i,0} : i \in \mathfrak{N}\}$ is generated by independent randomized observations on the distribution $\mathbf{F}$, then (A2) holds with probability one by the Strong Law of Large Numbers.

2.2. Noise Traders. We assume the noise trader order-flow process is a bivariate shot-noise process, and satisfies the SDE

$$(2.7) \quad d\mu_t = -\kappa \mu_t dt + \eta^+ dJ_t^+ - \eta^- dJ_t^-.$$

Here, κ is the mean-reverting rate, $\eta^\pm$ is the jump size, and $J^\pm = (J_t^\pm)_{t \in \mathfrak{T}}$ are Poisson processes (independent of all other processes) with intensity $\lambda^\pm$. For notational simplicity, and to ensure the market has no overall trend, we assume $\eta^+ = \eta^-$ and $\lambda^+ = \lambda^-$. Extending the analysis to asymmetric parameters is straightforward. Much of our results can be generalized beyond this specific model, however, using this specific model allows us to obtain explicit results.

2.3. Fundamental Price Dynamics. As found in several studies, backed by empirical findings (see e.g., [20]), we assume the fundamental asset price F^ν satisfies

$$(2.8) \quad dF_t^\nu = \left(\lambda_0 \nu_{0,t} + \lambda \nu_t^{(N)} + \lambda_\mu \mu_t \right) dt + dM_t, \quad t \in \mathfrak{T},$$

where F_0^ν is given, $M = (M_t)_{t \in \mathfrak{T}}$ is a martingale (independent of initial inventories), $\nu^{(N)} = (\nu_t^{(N)})_{t \in \mathfrak{T}}$ is the average trading of all minor agents so that

$$\nu_t^{(N)} := \frac{1}{N} \sum_{i \in \mathfrak{N}} \nu_{i,t},$$

$\mu = (\mu_t)_{t \in \mathfrak{T}}$ is the order-flow of noise traders, and $\lambda_0, \lambda, \lambda_\mu \geq 0$ are the strength of the (linear) major agent's, minor agents', and noise traders' permanent impact, respectively, on the fundamental price – if order-flow is positive (more buyers than sellers), then prices are pushed upwards, if it is negative (more sellers than buyers), then prices are pushed downwards. The martingale term models the aggregate random impact of cancellations and posts in the limit order book on the fundamental price dynamics (see Fig. 1).

Here, the major and minor agents are coupled via the fundamental asset's drift in (2.8). In the large N population limit of HFTs, each individual minor agent has an asymptotically negligible impact on the fundamental price. The aggregated effect of all HFTs, however, is non-negligible.

2.4. The Agents' Performance Criterion. The major agent's objective is to maximize her finite-time horizon wealth, subject to an inventory penalty. Specifically, her performance criterion is

$$(2.9) \quad J_0(\nu_0; \nu_{-0}) := \mathbb{E} \left[X_{0,T}^{\nu_0} + Q_{0,T}^{\nu_0} (F_T^\nu - \alpha Q_{0,T}^{\nu_0}) - \phi \int_0^T (Q_{0,s}^{\nu_0})^2 ds \right],$$

where $\nu_{-0} := (\nu_i)_{i \in \mathfrak{N}}$ are the minor agents' trading strategies, and α and ϕ are positive scalars.

In (2.9), the term $Q_{0,T}^{\nu_0} (F_T^\nu - \alpha Q_{0,T}^{\nu_0})$ corresponds to the liquidation value of assets remaining at the end of the trading horizon. The quadratic term acts to penalize non-zero terminal inventories, hence, as $\alpha \rightarrow \infty$, the optimal strategy forces zero inventories at the end of the trading horizon. The running penalty term $\phi \int_0^T (Q_{0,s}^{\nu_0})^2 ds$ acts as an urgency penalization. When ϕ is large, the agent attempts to liquidate inventories quickly, while when $\phi = 0$, the agent is indifferent about holding inventories through time. [16] demonstrate that such penalties may be interpreted as stemming from the agent's aversion to model uncertainty (also known as ambiguity aversion).

In the sequel, we aim to maximize the performance criterion over a class of admissible trading strategies $\mathcal{A}$ defined as

$$\mathcal{A} := \left\{ \nu \mid \nu \text{ is } \mathcal{G}\text{-adapted \& } \mathbb{E} \left[\int_0^T \nu_t^2 dt \right] < \infty \right\},$$

where $\mathcal{G} = (\mathcal{G}_t)_{t \in \mathfrak{T}}$ and $\mathcal{G}_t = \mathcal{F}_t \vee \sigma((Q_{i,0})_{i \in \mathfrak{N}})$, i.e., the filtration $\mathcal{G}$ is the filtration $\mathcal{F}$ expanded by the inventories of all agents.

A minor agents' performance criterion is

$$(2.10) \quad J_i(\nu_i; \nu_{-i}) := \mathbb{E} \left[X_{i,T}^{\nu_i} + Q_{i,T}^{\nu_i} (F_T^\nu - \psi Q_{i,T}^{\nu_i}) \right], \quad i \in \mathfrak{N},$$

where $\nu_{-i} := (\nu_0, \nu_1, \dots, \nu_{i-1}, \nu_{i+1}, \dots, \nu_N)$ is the collection of trading rates excluding minor agent- i , and $Q_{i,T}^{\nu_i} (F_T^\nu - \psi Q_{i,T}^{\nu_i})$ is the value of closing her position at the end of the trading

horizon, with the penalty parameter $\psi > 0$. Analogous to the major agent, we take the minor agent's set of admissible strategies to be $\mathcal{A}$. [9] study a sea of only minor-agents (in the sense that each agent impacts the price with a factor of N^{-1}) but they also include a risk term as in (2.9), we could easily include such a term in (2.10).

We refer to the case where minor agents and noise traders do not trade, i.e., when $\mu_t = 0$ and $\nu_{i,t} = 0$, $i \in \mathfrak{N}$, as the Almgren-Chriss (AC) framework (see Appendix D).

3. A Mean-Field Game Approach. The major-minor agent optimal execution problem is to maximize (2.9)-(2.10), and may be seen as a stochastic game where coupling stems from the fundamental asset price dynamics (2.8). In this formulation, however, the control input of each individual agent depends on the control of all other agents, and those are unknown a priori. The problem, therefore, falls within a class of stochastic dynamic games with mean-field couplings in the fundamental asset price dynamics. This problem is intractable for a large N population of minor agents, as it is not feasible for each individual agent to collect detailed state information of all the other agents. To deal with this, we develop a MFG framework through the following steps:

1. *Infinite population limit of minor agents:* In this step we establish a Nash equilibrium, characterized by an equilibrium relationship between the individual strategies for major and minor agents and the overall trading effect of minor agents on a given agent through the fundamental asset price dynamics. This equilibrium relationship is described by a MFG consistency condition. To this end, we define the minor agents mean trading speed as $\bar{\nu} = (\bar{\nu}_t)_{t \in \mathfrak{T}}$, where

$$(3.1) \quad \bar{\nu}_t = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \nu_{i,t},$$

and assume this limit exists, and the minor agents corresponding mean inventory $\bar{Q} = (Q_t^{\bar{\nu}})_{t \in \mathfrak{T}}$, where

$$(3.2) \quad Q_t^{\bar{\nu}} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N Q_{i,t}^{\nu_i}.$$

2. *ϵ -Nash equilibrium for the finite population of minor agents:* In this step, we show the set of decentralized MFG trading strategies (developed in Step 1) establishes an ϵ_N -Nash equilibrium for a finite N population of minor agents where ϵ_N goes to zero asymptotically as N approaches infinity.

In the remainder of this section we proceed to demonstrate the details of Step 1. First, we approximate the average trading of minor agents $\nu^{(N)}$ in the population limit (i.e. $N \rightarrow \infty$) by a process $\bar{\nu}$ with the postulated form

$$(3.3) \quad \bar{\nu}_t = g_0(t) + g_1(t) Q_{0,t}^{\nu_0} + g_2(t) Q_t^{\bar{\nu}} + g_3(t) \mu_t,$$

where $\{g_0, \dots, g_3\}$ are (continuous) deterministic functions of time to be characterized later via self consistency equations in Section 4.1. The ansatz in (3.3) is similar to, but different from, what is used in linear-quadratic problems. It is motivated by the optimal strategies

form in [20], where the authors find that (in a single agent setting) the optimal strategy in the presence of noise traders is of the form $g_0(t) + g_1(t)q_t + g_2(t)\mu_t$. From the structure of our major-minor agent problem, we expect the major and minor agents' inventories to affect the mean-field.

Second, we formulate and solve two auxiliary stochastic optimal control problems for the major agent and a representative minor agent by replacing the finite population process $\nu^{(N)}$ in (2.8) with its mean-field $\bar{\nu}$. Finally, we present consistency requirements for the mean-field approximation, such that the average trading rate of minor agents obtained from auxiliary stochastic optimal control problems coincides in the population limit with $\bar{\nu}$ assumed in (3.3). This is the MFG consistency condition.

3.1. Infinite Number of Minor Agents. *Problem I: Major Agent's Auxiliary Stochastic Optimal Control Problem.* By replacing the process $\nu^{(N)}$ in (2.8) with $\bar{\nu}$ given in (3.3), we obtain the following modified dynamics

$$(3.4a) \quad dQ_{0,t}^{\nu_0} = \nu_{0,t} dt, \quad Q_{0,0}^{\nu_0} = Q_{0,0},$$

$$(3.4b) \quad dQ_t^{\bar{\nu}} = \bar{\nu}_t dt, \quad Q_{0,0}^{\bar{\nu}} = Q_0,$$

$$(3.4c) \quad \begin{aligned} d\bar{F}_t^{\nu_0, \bar{\nu}} &= \left(\lambda_0 \nu_{0,t} + \lambda(g_0(t) + g_1(t) Q_{0,t}^{\nu_0} + g_2(t) Q_t^{\bar{\nu}}) + (\lambda_\mu + \lambda g_3(t)) \mu_t \right) dt + dM_t, \\ \bar{F}_0^{\nu_0, \bar{\nu}} &= F_0, \end{aligned}$$

$$(3.4d) \quad d\bar{X}_{0,t}^{\nu_0} = -(\bar{F}_t^{\nu_0, \bar{\nu}} + a_0 \nu_{0,t}) \nu_{0,t} dt, \quad \bar{X}_{0,0} = x_0,$$

where the fundamental asset price process has been modified to $\bar{F}^{\nu_0, \bar{\nu}} = (\bar{F}_t^{\nu_0, \bar{\nu}})_{t \in \mathfrak{T}}$ (which is the dynamics in (2.8) where $\nu_t^{(N)}$ is replaced by the mean-field $\bar{\nu}_t$ in (3.3)), $\bar{X}^{\nu_0} = (\bar{X}_t^{\nu_0})_{t \in \mathfrak{T}}$ is the major agent's corresponding cash process, and the major agent's auxiliary performance criterion becomes

$$(3.4e) \quad \bar{J}_0(\nu_0, \bar{\nu}) := \mathbb{E} \left[\bar{X}_{0,T}^{\nu_0} + Q_{0,T}^{\nu_0} (\bar{F}_T^{\nu_0, \bar{\nu}} - \alpha Q_{0,T}^{\nu_0}) - \phi \int_0^T (Q_{0,s})^2 ds \right].$$

The major agent's auxiliary stochastic control problem is to obtain

$$(P1) \quad \bar{J}_0(\bar{\nu}) = \sup_{\nu_0 \in \mathcal{A}} \bar{J}_0(\nu_0, \bar{\nu}),$$

and if the supremum is attained in $\mathcal{A}$, then we denote the optimal strategy by ν_0^* . The solution of this stochastic optimal control problem is provided in Section 3.2.

Problem II: Representative Minor Agent's Auxiliary Stochastic Optimal Control Problem. We now present the problem for a representative minor agent, given the strategy of the major agent and other minor agents. By taking the mean-field limit of the problem, we replace both the process $\nu^{(N)}$, the empirical average trading speed, in (2.8) with $\bar{\nu}$, the mean-field trading speed of all minor agents, given in (3.3) and ν_0 by the optimal trading speed ν_0^* from Problem

(P1). We obtain the following we obtain the following modified dynamics

$$(3.5a) \quad dQ_{i,t}^{\nu_i} = \nu_{i,t} dt, \quad Q_{i,0}^{\nu_i} = Q_{i,0},$$

$$(3.5b) \quad d\bar{F}_t^{\nu_0^*, \bar{\nu}} = \left(\lambda_0 \nu_{0,t} + \lambda(g_0(t) + g_1(t) Q_{0,t}^{\nu_0^*} + g_2(t) \mathcal{Q}_t^{\bar{\nu}}) + (\lambda_\mu + \lambda g_3(t)) \mu_t \right) dt + dM_t, \\ \bar{F}_0 = F_0,$$

$$(3.5c) \quad d\bar{X}_{i,t}^{\nu_i} = -(\bar{F}_t^{\nu_0^*, \bar{\nu}} + a \nu_{i,t}) \nu_{i,t} dt, \quad \bar{X}_{i,0}^{\nu_i} = x_i,$$

where $\bar{X}_i^{\nu_i} = ((\bar{X}_{i,t}^{\nu_i})_{t \in \mathfrak{T}})$ are the minor agents' corresponding cash processes, and the minor agent's auxiliary performance criterion is

$$(3.5d) \quad \bar{J}_i(\nu_i, \bar{\nu}) := \mathbb{E} \left[\bar{X}_{i,T}^{\nu_i} + Q_{i,T}^{\nu_i} (\bar{F}_T^{\nu_0^*, \bar{\nu}} - \psi Q_{i,T}^{\nu_i}) \right].$$

Here, the asset price process is evaluated along the major agent's optimal response to the mean-field, hence $\bar{F}_t^{\nu_0^*, \bar{\nu}}$ appears in the performance criterion and cash-process. We suppress the dependence on ν_0^* in the performance criteria (3.5c), as ν_0^* is given in terms of the mean-field $\bar{\nu}$ through the solution of Problem (P1).

Minor agent- i 's auxiliary control problem is to obtain

$$(P2) \quad \bar{J}_i(\bar{\nu}) = \sup_{\nu_i \in \mathcal{A}} \bar{J}_i(\nu_i, \bar{\nu}),$$

and if the supremum is attained in $\mathcal{A}$, then we denote the optimal strategy by ν_i^* . The solution of this stochastic optimal control problem is provided in Section 3.2.

3.2. Solutions to Limiting Stochastic Optimal Control Problems for Major and Minor Agents. This section presents solutions to the auxiliary stochastic optimal control problems (3.4) and (3.5) using dynamic programming principle.

Solution to Problem I. This problem falls within the class of standard optimal execution problems of a single agent with a time dependent drift in the fundamental price and time dependent permanent impact. For $x, f \in \mathbb{R}$, $q \in \mathbb{R}_+$ and $t \in \mathfrak{T}$, we define the performance criteria under the arbitrary trading strategy $\nu_0 \in \mathcal{A}$ as

$$H_0^{\nu_0, \bar{\nu}}(t, x, f, q, y, u) = \mathbb{E}_{t,x,f,q,y,u} \left[\bar{X}_{0,T}^{\nu_0} + Q_{0,T}^{\nu_0} (\bar{F}_T^{\nu_0, \bar{\nu}} - \alpha Q_{0,T}^{\nu_0}) - \phi \int_t^T (Q_{0,s}^{\nu_0})^2 ds \right]$$

where $\mathbb{E}_{t,x,f,q,y,u}[\cdot] := \mathbb{E}[\cdot \mid \bar{X}_{0,t}^{\nu_0} = x, \bar{F}_t^{\nu_0, \bar{\nu}} = f, Q_{0,t}^{\nu_0} = q, \mathcal{Q}_t^{\bar{\nu}} = y, \mu_t = u]$. The value function is then

$$(3.6) \quad H_0^{\bar{\nu}}(t, x, f, q, y, u) := \sup_{\nu_0 \in \mathcal{A}} H_0^{\nu_0, \bar{\nu}}(t, x, f, q, y, u).$$

Here, we use the notation H to denote the value function, rather than J as in the previous section, to highlight the explicit dependence on the state variables.

Theorem 3.1. Assume that the value function $H_0(t, x, f, q, y, u)$ for the stochastic optimal control problem (3.4) is a C^1 function of the variables t, x, q, y and u , and C^2 in f . Then, the major agent's auxiliary value function (3.6) admits the ansatz

$$(3.7) \quad H_0^\bar{\nu}(t, x, f, q, y, u) = x + q f + h_0(t, y, u) + h_1(t, y, u) q + h_2(t) q^2.$$

where the functions h_0, h_1 , and h_2 satisfy the coupled PDEs

$$(3.8a) \quad \left(\partial_t + \lambda(g_0(t) + g_2(t)y + g_3(t)u) \partial_y + \mathcal{L}^u \right) h_0 + \frac{1}{4a_0} (h_1)^2 = 0,$$

$$(3.8b) \quad \left(\partial_t + \lambda(g_0(t) + g_2(t)y + g_3(t)u) \partial_y + \mathcal{L}^u \right) h_1 + \lambda g_1(t) \partial_y h_0 \\ + (\lambda g_0(t) + \lambda g_2(t)y + \lambda g_3(t)u + \lambda_\mu u) + \frac{1}{2a_0} (\lambda_0 + 2h_2) h_1 = 0,$$

$$(3.8c) \quad \partial_t h_2(t) + \lambda g_1(t) \partial_y h_1 + \lambda g_1(t) + \frac{1}{4a_0} (\lambda_0 + 2h_2)^2 - \phi = 0,$$

with terminal conditions

$$h_0(T, y, u) = h_1(T, y, u) = 0, \quad \text{and} \quad h_2(T, y, u) = -\alpha.$$

Moreover, the optimal trading rate in the major agent's auxiliary control problem is

$$(3.9) \quad \nu_{0,t}^* = \frac{1}{2a_0} \left[(\lambda_0 + 2h_2(t)) Q_{0,t}^{\nu_0} + h_1(t, \mathcal{Q}_t^\bar{\nu}, \mu_t) \right].$$

Proof. See A.1 ■

Solution to Problem II. This problem falls within the class of optimal execution problems for a single agent with a time-dependent fundamental price drift, and zero permanent impact (in contrast to the auxiliary control problem for the major which has permanent impact). For $x, f, q \in \mathbb{R}$ and $t \in \mathfrak{T}$, we define the performance criterion for a generic minor agent $i = 1, \dots, N$ under the arbitrary trading strategy $\nu_i \in \mathcal{A}$ as

$$H_i^{\nu_i, \bar{\nu}}(t, x, f, q_i, q, y, u) = \mathbb{E}_{t, x, f, q_i, q, y, u} \left[\bar{X}_{i,T}^{\nu_i} + Q_{i,T}^{\nu_i} (\bar{F}_T^{\nu_0^*, \bar{\nu}} - \psi Q_{i,T}^{\nu_i}) \right],$$

where $\mathbb{E}_{t, x, f, q_i, q, y, u}[\cdot] := \mathbb{E}[\cdot \mid \bar{X}_{i,t}^{\nu_i} = x, \bar{F}_t^{\nu_0^*, \bar{\nu}} = f, Q_{i,t}^{\nu_i} = q_i, Q_{0,t}^{\nu_0^*} = q, \mathcal{Q}^\bar{\nu} = y, \mu_t = u]$ and her value function is

$$(3.10) \quad H_i^\bar{\nu}(t, x, f, q_i, q, y, u) := \sup_{\nu_i \in \mathcal{A}} H_i^{\nu_i, \bar{\nu}}(t, x, f, q_i, q, y, u).$$

Theorem 3.2. Assume that the value function $H_i(t, x, f, q_i, q, y, u)$ for the auxiliary stochastic optimal control problem (3.5) is a C^1 function of the variables t, x, q, q_i, y and u , and C^2 function of variable f . Then, a generic minor agent's auxiliary value function (3.10) admits the ansatz

$$H_i^\bar{\nu}(t, x, f, q_i, q, y, u) = x + f q_i + l_0(t, q, y, u) + l_1(t, q, y, u) q_i - \frac{a}{\frac{a}{\psi} + T} q_i^2,$$

where l_0 and l_1 are deterministic functions of time (t), the major agent's inventory (q), the minor agents mean inventory (y) and noise traders speed of trading (u). The optimal trading rate $\{\nu_i^*\}_{i \in \mathfrak{N}}$ is

$$(3.11) \quad \nu_{i,t}^* = - \left(\frac{a}{\psi} + (T - t) \right)^{-1} Q_{i,t}^{\nu_i^*} + \frac{1}{2a} l_1 \left(t, Q_{0,t}^{\nu_0^*}, \mathcal{Q}^\bar{\nu}, \mu_t \right).$$

Proof. See A.2 ■

3.3. Verification. The control derived from the HJB equations are stochastic in general, due to the solutions ν_0^*, ν_i^* adapted nature. We make a further ansatz

$$\begin{aligned} h_0(t, y, u) &= h_0^0(t) + h_0^1(t) y + h_0^2(t) y^2 + h_0^\mu(t) u y + h_0^{0,\mu} u + h_0^{2,\mu} u^2, \\ h_1(t, y, u) &= h_1^0(t) + h_1^1(t) y + h_1^\mu(t) u, \end{aligned}$$

and

$$\begin{aligned} l_0(t, q, y, u) &= l_0^0(t) + l_0^{1,q}(t) q + l_0^{2,q}(t) q^2 + l_0^{1,y}(t) y + l_0^{2,y}(t) y^2 + l_0^{0,\mu} u + l_0^{2,\mu} u^2, \\ l_1(t, q, y, u) &= l_1^0(t) + l_1^1(t) q + l_1^2(t) y + l_1^\mu(t) u, \end{aligned}$$

where $l_0(t, q, y, u)$ is a quadratic function of q, y, u . This leads to the ODEs (A.5), (A.10) in A.2 which are at most quadratic in the state variables. Therefore, we have that if the system of coupled ODEs admits a solution, then

- (i) for each t , the functions $H_0(t, x, f, q, y, u), H_i(t, x, f, q_i, q, y, u)$ are a C^2 map into $\mathbb{R}$,
- (ii) the partial derivatives of $H_0(t, x, f, q, y, u), H_i(t, x, f, q_i, q, y, u)$ are continuous.

As the minor agents' strategies are $\mathcal{F}_t$ -predictable, and $\bar{\nu}$ is linear in the state variables, the verification theorem then holds by an application of Theorem 9.1 in [41], implying that

$$(3.12) \quad J_0(\nu_0^*, \nu_{-0}^*) = \sup_{\nu^0} J_0(\nu_0, \nu_{-0}^*) \quad \text{and} \quad J_i(\nu_i^*, \nu_{-i}^*) = \sup_{\nu^i} J_i(\nu_i, \nu_{-i}^*)$$

It is worth point out that while in our approach we have a common noise component, it poses no problems in our analysis due to the way the agents' actions feed through the dynamics and the specific performance criterion the agents use. If the common noise component feed through in a non-linear fashion, then there would be additional complexities, and perhaps the problem would then be intractable.

4. Fixed Point Equations.

4.1. Mean-Field Game Fixed Point. In this section, we study the consistency of the mean-field. Specifically, when the MFG strategies major-minor agents are employed, the resulting mean-field for the minor agents must be consistent with the assumption in (3.3). By averaging over all minor-agents, and using the form of l_1 provided in (A.10), we obtain

$$\lim_{n \rightarrow \infty} \frac{1}{N} \sum_{i \in \mathfrak{N}} \nu_{i,t}^* = - \left(\frac{a}{\psi} + (T - t) \right)^{-1} Q_{i,t}^{\nu_i^*} + \frac{1}{2a} \left(l_1^0(t) + l_1^1 Q_{0,t}^{\nu_0^*} + l_1^2 Q_{i,t}^{\bar{\nu}} + l_1^\mu(t) \mu_t \right).$$

Imposing the consistency condition $\bar{\nu}_t = \lim_{n \rightarrow \infty} \frac{1}{N} \sum_{i \in \mathfrak{N}} \nu_{i,t}^*$, by comparing (3.3) with the above, yields the following set of consistency equations:

$$(4.1a) \quad g_0(t) = \frac{1}{2a} l_1^0(t),$$

$$(4.1b) \quad g_1(t) = \frac{1}{2a} l_1^1(t),$$

$$(4.1c) \quad g_2(t) = - \left(\frac{a}{\psi} + (T - t) \right)^{-1} + \frac{1}{2a} l_1^2(t),$$

$$(4.1d) \quad g_3(t) = \frac{1}{2a} l_1^\mu(t).$$

By combining (4.1) with (A.5) and (A.10), we obtain the major and minor agents' optimal controls through the solution of a system of coupled Riccati ODEs. We denote the corresponding mean-field to this fixed point as $\bar{\nu}^*$.

4.2. Behaviour of the Mean-Field Game System with Infinite Penalty. In this section, we consider the trading strategy for the major and minor agents when the minor agents have infinite temporary impact, (i.e., as $a \rightarrow \infty$). In this limit, we anticipate that the minor agents do not trade, as trading has an infinite cost.

Lemma 4.1. *In the limit, as $a \rightarrow \infty$, we have $\nu_i^* \rightarrow 0$.*

Proof. Given a solution to (A.10) with $a \rightarrow \infty$ when substituting (4.1) and $g_i(t) = 0$ for $i = 0, 1, 2, 3$ into the ODEs, for any bounded solutions $l_1^0, l_1^1, l_1^2, l_1^\mu$, (4.1) implies that $g_i(t) = 0$ for $i = 0, 1, 2, 3$ as $a \rightarrow \infty$, which shows the consistency equations are satisfied. ■

As a result, the mean-field vanishes, and the major agent's optimal strategy is the solution to the PDEs

$$(4.2a) \quad (\partial_t + \mathcal{L}^\mu) h_0(t, y, u) + \frac{1}{4a_0} h_1(t, y, u)^2 = 0, \quad h_0(T, y, u) = 0,$$

$$(4.2b) \quad (\partial_t + \mathcal{L}^\mu) h_1(t, y, u) + \lambda \lambda_\mu u + \frac{1}{2a_0} (\lambda_0 + 2h_2(t)) h_1(t, y, u) = 0, \quad h_1(T, y, u) = 0,$$

$$(4.2c) \quad \partial_t h_2(t) + \frac{1}{4a_0} (\lambda_0 + 2h_2(t))^2 - \phi = 0, \quad h_2(T) = -\alpha.$$

These PDEs are identical to the equations found in [20], where the authors study the effect of order-flow in an optimal execution problem for a single agent.

5. Decentralized mean-field Game Trading Strategies: The ϵ_N -Nash Equilibrium Property. In this section, we prove the decentralized mean-field solution presented in the previous section has the ϵ_N -Nash equilibrium property. We study this in the absence of noise traders. This assumption simplifies the analysis, as the strategies of the major and minor agents are deterministic when noise traders are not present. In the presence of noise traders, the main difficulty lies in finding stochastic bounds which are as tight as the deterministic ones used in the proof. To this end, we make the following assumption.

(A3) Assume there exists a unique solution set (g_0, g_1, g_2) in $\mathcal{C}_b(\mathbb{R}_+)$ to the MFG system (4.1).

Under (A3), the MFG trading strategies for a finite population of major and minor agents are

$$(5.1a) \quad \nu_{0,t}^* = \frac{1}{2a_0} (\lambda_0 + 2h_2(t)) Q_{0,t}^{\nu_0^*} + \frac{1}{2a_0} h_1(t, Q_{0,t}^{\nu_0^*}, \mathcal{Q}^{\bar{\nu}^*}), \quad \text{and}$$

$$(5.1b) \quad \nu_{i,t}^* = -\frac{\psi}{a+\psi(T-t)} Q_{i,t}^{\nu_i^*} + \frac{1}{2a} l_1(t, Q_{0,t}^{\nu_0^*}, \mathcal{Q}^{\bar{\nu}^*}), \quad i \in \{1, \dots, N\},$$

where $h_1(\cdot)$ and $h_2(\cdot)$ are determined by (A.5), and $l_1(\cdot)$ is given in (A.10), with $g_3(t) = 0$ for all $t \in \mathfrak{T}$ as there are no noise traders. In equations (A.5) and (A.10), the functions g_0 , g_1 , and g_2 are the unique solution of the MFG system (4.1). The implementation of the overall MFG trading rates for the major and minor agents (5.1) have a decentralized form. That is, each agent enacts a control law consisting of the feedback of its own quantity of shares plus a pre-computed deterministic effect due to the cross interaction of major-minor agents.

Applying the optimal trading strategies (5.1) in a finite population of minor agents (2.1)-(2.6), and letting $\boldsymbol{\nu}^* = (\nu_0^*, \{\nu_i^*\}_{i \in \mathfrak{N}})$, yields the following for the major agent

$$(5.2a) \quad dQ_{0,t}^{\nu_0^*} = \nu_{0,t}^* dt, \quad Q_{0,0}^{\nu_0^*} = Q_{0,0},$$

$$(5.2b) \quad dX_{0,t}^{\nu_0^*} = -(F_t^{\nu^*} + a_0 \nu_{0,t}^*) \nu_{0,t}^* dt, \quad X_{0,0}^{\nu_0^*} = x_0,$$

and for each minor-agent $i \in \mathfrak{N}$,

$$(5.2c) \quad dQ_{i,t}^{\nu_i^*} = \nu_{i,t}^* dt, \quad Q_{i,0}^{\nu_i^*} = Q_{i,0},$$

$$(5.2d) \quad dX_{i,t}^{\nu_i^*} = -(F_t^{\nu^*} + a \nu_{i,t}^*) \nu_{i,t}^* dt, \quad X_{i,0}^{\nu_i^*} = x_i,$$

where the agents optimal behavior are coupled through the optimal trading strategies in (5.1) and the fundamental asset price dynamics

$$(5.3) \quad dF_t^{\nu^*} = \left(\lambda_0 \nu_{0,t}^* + \frac{\lambda}{N} \sum_{i=1}^N \nu_{i,t}^* \right) dt + dM_t, \quad F_t^{\nu^*} = F_0.$$

To simplify notation, we often write F^* , X_i^* , Q_i^* , and $\mathcal{Q}^*$ as short-hand for F^{ν^*} , $X_i^{\nu_i^*}$, $Q_i^{\nu_i^*}$, and $\mathcal{Q}^{\nu^*}$.

Definition 5.1. *Given $\epsilon > 0$, the set of trading rates $\{\nu_i^* \in \mathcal{A} : i \in \mathfrak{N} \cup \{0\}\}$ for $N + 1$ major and minor agents generates an ϵ -Nash equilibrium with respect to the utility functions $\{J_i : i \in \mathfrak{N} \cup \{0\}\}$, if*

$$J_i(\nu_i^*, \nu_{-i}^*) \leq \sup_{\nu_i \in \mathcal{A}} J_i(\nu_i, \nu_{-i}^*) \leq J_i(\nu_i^*, \nu_{-i}^*) + \epsilon,$$

for any $i \in \mathfrak{N} \cup \{0\}$.

The ϵ -Nash equilibrium property of the trading rates $\{\nu_i^* \in \mathcal{A} : i \in \mathfrak{N} \cup \{0\}\}$ implies that for any given $\epsilon > 0$, there exists M_ϵ such that for any population size of minor agents $N \geq M_\epsilon$, any agent i within the population may only benefit ϵ by unilaterally (i.e., while any other agent $j \in \mathfrak{N} \cup \{0\}$, $j \neq i$, employs ν_j^*) deviating from its trading strategy ν_i^* .

We denote

$$(5.4) \quad \epsilon_N := \left| \frac{1}{N} \sum_{i=1}^N Q_{i,0} - \mathcal{Q}_0 \right| \equiv \left| \int_{\mathbb{R}} q d\mathbf{F}_N(q) - \int_{\mathbb{R}} q d\mathbf{F}(q) \right|,$$

where $\mathcal{Q}_0$ is the average of the distribution employed in the MFG, where the distributions $\mathbf{F}_N$ and $\mathbf{F}$ are defined in Assumption (A2). It is evident that under Assumption (A2) $\lim_{N \rightarrow \infty} \epsilon_N = 0$.

To facilitate the estimate of the bounds on the approximate finite game solution, we construct several auxiliary processes. First, introduce the auxiliary cash dynamics $\bar{X}_{-i}^* = (\bar{X}_{-i,t}^*)_{t \in \mathfrak{T}}$ for the N minor agents, where the fundamental price follows the response to the optimal mean-field strategy, but the cash accumulates according to an arbitrary strategy from agent- i . Specifically, $\bar{X}_{-i}^*$ satisfy the SDE

$$(5.5) \quad d\bar{X}_{-i,t}^* = -(\bar{F}_t^* + a \nu_{i,t}) \nu_{i,t} dt, \quad \bar{X}_{-i,0}^* = x_{i,0}, \quad i \in \mathfrak{N},$$

for an arbitrary $\nu_i \in \mathcal{A}$, and where $\bar{F}^* = (\bar{F}_t^*)_{t \in \mathfrak{T}}$ is the response of the fundamental price to the mean-field strategy $\bar{\nu}_t^* := g_0(t) + g_1(t) Q_{0,t}^* + g_2(t) \mathcal{Q}_t^*$, so that $\bar{F}^*$ satisfies the SDE

$$(5.6) \quad d\bar{F}_t^* = \{(\lambda_0 \nu_{0,t}^* + \lambda (g_0(t) + g_1(t) Q_{0,t}^* + g_2(t) \mathcal{Q}_t^*))\} dt + dM_t, \quad \bar{F}_0^* = F_0.$$

In (5.6) the infinite population mean-field process appears and not the finite population average of the minor agents' mean-field controls $\frac{1}{N} \sum_{i=1}^N \nu_{i,t}^*$ in (5.3).

Next, we introduce the cash processes subject to the optimal mean-field strategies $\bar{X}_i^* = (\bar{X}_{i,t}^*)_{t \in \mathfrak{T}}$ which satisfy the SDEs

$$(5.7) \quad d\bar{X}_{0,t}^* = -(\bar{F}_t^* + a_0 \nu_{0,t}^*) \nu_{0,t}^* dt, \quad \bar{X}_{0,0}^* = x_0,$$

$$(5.8) \quad d\bar{X}_{i,t}^* = -(\bar{F}_t^* + a \nu_{i,t}^*) \nu_{i,t}^* dt, \quad \bar{X}_{i,0}^* = x_i, \quad i \in \mathfrak{N}.$$

The difference between $\bar{X}_{-i}^*$ and $\bar{X}_i^*$ is that $\bar{X}_{-i}^*$ uses the arbitrary strategy $\nu_i \in \mathcal{A}$, while $\bar{X}_i^*$ uses the optimal mean-field strategy ν_i^* . In both cases, however, the fundamental price is the one generated by the mean-field strategy.

Continuing with defining additional auxiliary process, for a given $\nu_0 \in \mathcal{A}$, we let $\bar{F}_{-0}^* = (\bar{F}_{-0,t}^*)_{t \in \mathfrak{T}}$ denote the response of the fundamental price to the mean-field of the minor agents, but the major agent acting with the arbitrary strategy ν_0 . We further let $\bar{X}_{-0}^* = (\bar{X}_{-0,t}^*)_{t \in \mathfrak{T}}$ denote the major agent's cash process in response to this fundamental price, with the same arbitrary strategy ν_0 . In all, they satisfy the SDEs

$$(5.9) \quad \begin{aligned} d\bar{F}_{-0,t}^* &= (\lambda_0 \nu_{0,t} + \lambda \bar{\nu}_t^*) dt + dM_t \\ &= (\lambda_0 \nu_{0,t} + \lambda (g_0(t) + g_1(t) Q_{0,t}^* + g_2(t) Q_t)) dt + dM_t, \end{aligned} \quad \bar{F}_{-0,0}^* = F_0,$$

$$(5.10) \quad d\bar{X}_{-0,t}^* = -(\bar{F}_{-0,t}^* + a_0 \nu_{0,t}) \nu_{0,t} dt, \quad \bar{X}_{-0,0}^* = x_0.$$

Finally, we introduce the auxiliary fundamental price dynamics and cash processes in response to the finite population actions. To distinguish them in notation from the above, there are no overlines on the variables. So that $F_{-i}^* = (F_{-i,t}^*)_{t \in \mathfrak{T}}$, for $i \in \mathfrak{N} \cup \{0\}$, denotes the fundamental price in response to finite population action of all agents, excluding agent- i who acts according to an arbitrary strategy, acting according to the mean-field strategies. Specifically, for given $\nu_0 \in \mathcal{A}$ and $\nu_i \in \mathcal{A}$, $i \in \mathfrak{N}$, we have

$$(5.11) \quad dF_{-0,t}^* = \left(\lambda_0 \nu_{0,t} + \frac{\lambda}{N} \sum_{i \in \mathfrak{N}} \nu_{i,t}^* \right) dt + dM_t, \quad F_{-0,0}^* = F_0$$

$$(5.12) \quad dF_{-i,t}^* = \left(\lambda_0 \nu_{0,t}^* + \frac{\lambda}{N} \sum_{j \in \mathfrak{N}, j \neq i} \nu_j^* + \frac{\lambda}{N} \nu_i \right) dt + dM_t, \quad F_{-i,0}^* = F_0.$$

Similarly, we let $X_{-i}^* = (X_{-i,t}^*)_{t \in \mathfrak{T}}$, for $i \in \mathfrak{N} \cup \{0\}$, denote agent- i 's cash process in response to finite population action of all agents, excluding minor agent- i who acts according to an arbitrary strategy, acting according to the mean-field strategies. Specifically, for $i \in \mathfrak{N}$ we have

$$(5.13) \quad dX_{-0,t}^* = -(F_{-0,t}^* + a_0 \nu_{0,t}) \nu_{0,t} dt, \quad X_{-0}^*(0) = x_0,$$

$$(5.14) \quad dX_{-i,t}^* = -(F_{-i,t}^* + a \nu_{i,t}) \nu_{i,t} dt, \quad X_{-i}^*(0) = x_i.$$

Given all of these auxiliary processes, we have the following bounds on the deviations of the various fundamental prices and cash processes². Firstly, there exist $C > 0$ such

²For simplicity, we assume $Q_{0,T} = Q_{i,T} = 0$, so that all agents liquidate by time T .

that if $\int_0^T (\nu_{0,t})^2 dt > C$, then $J(\nu_0) < J(\nu_0^*)$, as $J(\nu_0)$ is quadratic in ν_0 and it may be bounded by $J(\nu_0) \leq \mathbb{E}[F_0 Q_0 + F_0 \int_0^T |\nu_{0,t}| dt - a \int_0^T \nu_{0,t}^2 dt]$. The same holds for $\nu_{i,t}$ as $J(\nu_i) \leq \mathbb{E}[F_0 \int_0^T |\nu_{i,t}| dt - a \int_0^T \nu_{i,t}^2 dt]$. Therefore, any alternative strategies for the finite player game is also bounded. Analogously, we can bound the optimal mean-field strategies.

Lemma 5.1. *Assume (A1)-(A3) hold. Then, for any $t \in \mathfrak{T}$ we have*

$$(5.15a) \quad \mathbb{E}|F_{-i,t}^* - \bar{F}_t^*| = O(\epsilon_N) + O(N^{-1}),$$

$$(5.15b) \quad \mathbb{E}|X_{-i,t}^* - \bar{X}_{-i,t}^*| = O(\epsilon_N) + O(N^{-1}),$$

for all $i = 1, \dots, N$, and

$$(5.15c) \quad \mathbb{E}|F_{-0,t}^* - \bar{F}_{-0,t}^*| = O(\epsilon_N),$$

$$(5.15d) \quad \mathbb{E}|X_{-0,t}^* - \bar{X}_{-0,t}^*| = O(\epsilon_N),$$

where ϵ_N is defined in (5.4), and $\bar{X}_{-i}^*$, $\bar{F}^*$, $\bar{F}_{-0}^*$, $\bar{X}_{-0}^*$, F_{-0}^* , F_{-i}^* , X_{-0}^* and X_{-i}^* are given in (5.5)-(5.14), respectively.

Proof. Proof in B.1.

Armed with these bounds, we are now in a position to state the main result of this section.

Theorem 5.1. *Assume (A1)-(A3) hold. Then, the set of MFG trading strategies $\{\nu_i^* \in \mathcal{A} : i \in \mathfrak{N} \cup \{0\}\}$ in (5.1a)-(5.1b) generates an ϵ -Nash equilibrium, i.e., for any fixed $i \in \mathfrak{N} \cup \{0\}$ we have*

$$(5.16) \quad J_i(\nu_i^*, \nu_{-i}^*) \leq \sup_{\nu_i \in \mathcal{A}} J_i(\nu_i, \nu_{-i}^*) \leq J_i(\nu_i^*, \nu_{-i}^*) + O(\epsilon_N) + O(N^{-1}),$$

where ϵ_N is given in (5.4), and $\lim_{N \rightarrow \infty} \epsilon_N = 0$ (a.s.).

Proof. Proof in B.2.

6. Simulation. Model parameters are estimated from empirical data in order to simulate the impact of using the mean-field strategy vs the order-flow strategy (which does not account for minor HFTs). The parameters used in the simulation are calibrated to data and they may be found in Appendix C.

We benchmark the mean-field strategy described above against the order-flow strategy from [20]. We denote the mean-field strategy as MFG and the order-flow strategy as OF, and the corresponding cash process from each as X_{MFG} and X_{OF} , respectively. We compare the profit-and-loss (PnL) of the strategies in basis points through the following formula

$$(6.1) \quad PnL = \frac{X_{MFG} - X_{OF}}{X_{MFG}} \times 10^4,$$

and report the results in Table 1. As the table shows, the MFG strategy significantly outperforms the OF strategy across most assets in terms of mean and median savings, and the left tails are light.

Table 1
Mean-Field Game vs Order-Flow strategy (Executing 20% of Order Flow).

Ticker	Mean	StDev	Quantiles					$\mathbb{P}(MFG < OF)$
			5%	25%	50%	75%	95%	
EBAY	1.37	0.60	0.39	0.96	1.37	1.78	2.37	1%
INTC	1.69	0.71	0.53	1.21	1.70	2.17	2.87	1%
DOW	1.54	0.92	0.02	0.91	1.53	2.15	3.06	5%
CSCO	1.22	0.62	0.21	0.80	1.22	1.64	2.25	2%
COST	0.88	0.51	0.06	0.53	0.87	1.21	1.72	4%
AXP	0.73	0.64	-0.32	0.31	0.73	1.15	1.77	13%
AMAT	1.75	1.04	0.08	1.06	1.75	2.45	3.47	4%
PNC	0.62	0.55	-0.26	0.25	0.61	0.98	1.52	13%
AA	1.68	1.20	-0.24	0.86	1.66	2.47	3.71	8%
FMER	1.46	1.07	-0.23	0.73	1.41	2.14	3.31	8%
CNQR	0.60	0.73	-0.58	0.09	0.59	1.08	1.80	21%
ROC	0.17	0.29	-0.30	-0.02	0.17	0.37	0.66	28%
IMGN	2.07	1.86	-0.90	0.78	2.00	3.30	5.22	13%
DK	0.56	1.44	-1.74	-0.44	0.54	1.52	2.98	36%
ARCC	0.61	0.58	-0.27	0.21	0.57	0.97	1.64	14%
BXS	0.74	0.98	-0.78	0.06	0.70	1.36	2.45	23%
AYI	0.01	0.12	-0.18	-0.07	0.01	0.09	0.23	48%

6.1. Major and minor agents with same liquidation horizon. In Figure 2, we plot the major agent's inventory versus execution time. The top panel shows the major agent's inventory when she follows the optimal MFG strategy and when the major agent ignores the presence of the HFTs – this latter strategy corresponds to the order-flow (OF) strategy in [20]. The second panel shows the difference between the two strategies. In comparison to [20], the major agent initially speeds up trading anticipating that the HFTs will short the asset, as the major agent is selling shares which pushes prices down. The major agent then slows down trading and speeds up again towards the end of the trading horizon in anticipation of the HFTs unwinding their short position which pushes the prices upwards.

Figure 3 shows the corresponding comparison for the minor agents. The minor agents anticipate the liquidating major agent (whether the major agent uses the MFG or OF strategy) and therefore initially short the stock, pushing prices down. As they aim to end the trading period with zero inventory, they slowly unwind their short position by repurchasing the stock. They do so slowly as they anticipate the major agent's continual selling pressure. The plot also demonstrates how the HFTs create a larger short position when the major agent follows the OF strategy in comparisons to the MFG strategy.

6.2. Sensitivity Analysis. In this section, we carry out a sensitivity analysis on various parameters. We take the baseline inventory to be 10% of daily order-flow, and permanent and temporary impact are taken for INTC as reported in Table 4 in Appendix C.

6.2.1. Permanent Price Impact. Permanent price impact affects the strength of stock price changes in the presence of order-flow from major and minor agents and noise traders. An increase in permanent price impact provides a greater opportunity for minor agents (HFTs)

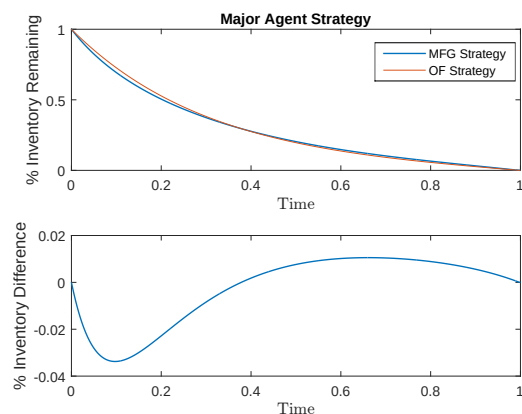


Figure 2. Optimal trading for major agent comparing (i) the MFG strategy with (ii) the optimal single agent strategy when the major agent ignores the HFTs (OF).



Figure 3. Optimal trading for minor agents comparing (i) the strategy they follow when the major agent accounts for minor agents, and (ii) the strategy they follow when the major agent ignores the optimal actions of the minor agents (OF).

to profit from statistical arbitrage strategies. The top panel of Figure 4 shows how the major agent's benefit from executing a MFG strategy increases with permanent price impact parameter.

6.2.2. Temporary Price Impact. Temporary price impact increases the immediate cost of executing large trades. An increase in temporary price impact deters all agents from trading too quickly. In particular, it penalizes the major agent from varying her trading speed too much, and incentivizes a more consistent trading speed. As shown earlier, the major agent's benefit in the MFG strategy stems from executing quickly early on, and then more slowly at other times, so this benefit is muted in the presence of a higher temporary price impact, as seen in the bottom panel of Figure 4.

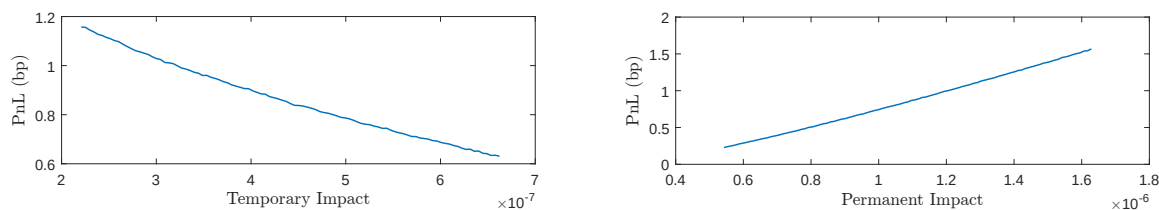


Figure 4. Temporary and permanent price impact sensitivity. Here, PnL is the mean relative savings of the MFG strategy over the OF strategy defined in (6.1).

6.2.3. Execution Quantity. Another factor that affects the major agent's performance is the quantity of shares she is executing. Tables 2 and 3 show the mean increase in relative performance for various levels of initial inventory when compared with the OF and AC strategies, respectively – using (6.1) using X_{OF} or X_{AC} . Across all of the assets, inventory

levels, the MFG strategy outperforms both OF and AC. When compared with OF (Table 2), the MFG strategy appears to perform better with increasing volume. This occurs since the MFG strategy accounts the actions of the minor agents in response to the increased major agent's volume, while OF ignores this. When compared with AC (Table 2), the MFG strategy appears to increase in initial inventory for half of the assets considered (from EBAY down to AMAT in the table). For the other half of the assets considered (from PNC onwards in the table), it is initially decreasing, but then increases once inventory is increased beyond around 10%. This trading behaviour is largely consistent with those found in [26]. As well, for each asset and inventory level, the mean relative performance of the MFG strategy over the AC is better than that over the OF strategy. This is expect since both the MFG and OF strategy account for noise traders, while the AC does not.

Table 2

Mean relative performance of MFG over OF strategy for various major agent initial inventory levels.

Ticker	Initial Inventory as % of Order Flow									
	2%	4%	6%	8%	10%	12%	14%	16%	18%	20%
EBAY	0.13	0.28	0.40	0.53	0.68	0.82	0.95	1.08	1.22	1.37
INTC	0.18	0.33	0.50	0.66	0.84	1.00	1.16	1.33	1.50	1.66
DOW	0.15	0.30	0.45	0.61	0.74	0.90	1.05	1.21	1.35	1.50
CSCO	0.13	0.23	0.36	0.48	0.60	0.72	0.85	0.96	1.08	1.19
COST	0.09	0.18	0.25	0.35	0.44	0.50	0.59	0.68	0.78	0.87
AXP	0.08	0.14	0.22	0.28	0.36	0.43	0.51	0.58	0.65	0.72
AMAT	0.17	0.35	0.53	0.71	0.87	1.04	1.21	1.39	1.56	1.74
PNC	0.06	0.12	0.20	0.24	0.30	0.36	0.43	0.50	0.55	0.62
AA	0.20	0.37	0.52	0.65	0.84	1.01	1.15	1.32	1.48	1.66
FMER	0.18	0.33	0.46	0.60	0.72	0.88	1.01	1.14	1.30	1.43
CNQR	0.10	0.12	0.19	0.24	0.30	0.35	0.41	0.46	0.51	0.58
ROC	0.02	0.03	0.05	0.07	0.08	0.10	0.12	0.13	0.15	0.17
IMGN	0.34	0.48	0.65	0.85	1.01	1.20	1.42	1.62	1.81	2.02
DK	0.07	0.13	0.17	0.23	0.28	0.35	0.35	0.45	0.54	0.57
ARCC	0.14	0.21	0.24	0.28	0.31	0.37	0.42	0.48	0.54	0.58
BXS	0.19	0.22	0.30	0.32	0.38	0.46	0.52	0.57	0.64	0.70
AYI	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01

7. Concluding Remarks. This work is the first to generalize, and completely solve, the classical optimal liquidation problem to a multiple agent setting. Here, in addition to a major agent who is liquidating a large portion of shares, there are a number of minor agents (high-frequency traders (HFTs)) who detect and trade along with the liquidator. Cross interaction between the minor and major agents occur through the impact that all traders have on the drift of the fundamental price. This formulation gives rise to a stochastic dynamic game with mean-field couplings in the fundamental price.

We demonstrate how this problem may be solved in the mean-field sense resulting in solving a deterministic fixed point problem. Under appropriate conditions, we demonstrated that a unique solution to the fixed point problem exists, and that the resulting decentralized trading strategies form an ϵ -Nash equilibrium for the finite population problem.

The general form of the mean-field optimal strategy is that the minor agents trade along

Table 3

Mean relative performance of MFG over AC strategy for various major agent initial inventory levels.

Ticker	Initial Inventory as % of Order Flow									
	2%	4%	6%	8%	10%	12%	14%	16%	18%	20%
EBAY	0.17	0.30	0.41	0.54	0.69	0.83	0.96	1.09	1.23	1.37
INTC	0.28	0.38	0.54	0.68	0.86	1.01	1.18	1.35	1.51	1.67
DOW	0.27	0.35	0.48	0.64	0.76	0.92	1.06	1.22	1.36	1.51
CSCO	0.28	0.31	0.42	0.52	0.63	0.74	0.87	0.98	1.10	1.21
COST	0.21	0.24	0.29	0.38	0.46	0.52	0.61	0.70	0.79	0.88
AXP	0.19	0.19	0.26	0.31	0.38	0.45	0.53	0.59	0.66	0.73
AMAT	0.51	0.52	0.64	0.79	0.94	1.10	1.26	1.43	1.60	1.77
PNC	0.22	0.20	0.25	0.28	0.33	0.38	0.45	0.52	0.56	0.63
AA	0.94	0.77	0.78	0.86	1.00	1.15	1.26	1.43	1.57	1.74
FMER	1.43	1.04	0.95	0.98	1.02	1.13	1.23	1.32	1.46	1.59
CNQR	0.69	0.44	0.40	0.39	0.43	0.46	0.50	0.54	0.58	0.64
ROC	0.28	0.16	0.14	0.13	0.13	0.15	0.15	0.16	0.18	0.19
IMGN	2.59	1.71	1.48	1.48	1.51	1.62	1.78	1.93	2.08	2.27
DK	0.95	0.59	0.49	0.47	0.46	0.51	0.49	0.56	0.64	0.66
ARCC	1.48	0.98	0.79	0.70	0.66	0.66	0.66	0.69	0.73	0.75
BXS	2.32	1.45	1.14	0.99	0.91	0.90	0.90	0.90	0.93	0.97
AYI	0.56	0.29	0.19	0.15	0.12	0.10	0.09	0.07	0.07	0.06

side the major agent, and attempt to gain from the price impact the major agent has on prices. The major agent anticipates this behaviour and speeds up trading near the beginning and end of the trading horizon to take advantage of the lower price impacts of the minor agents, while slowing down in the middle so that they do not execute too many stocks when the mid-price is low.

There are many directions open for future research, such as including trading constraints for the minor agents and solving for their optimal trading horizon. Another interesting problem would be to solve the problem where the minor agents and major agents do not have distributional information on the initial inventory. Clearly, in real markets agents are blind to other agent's position, but they may be able to sniff out a large trader liquidating a position. Knowing how to "hide" the major agent's strategy from the minor agents is an important problem that we aim to address in future work.

Appendix A. Proof of Value Functions. In this appendix, we interpret $\bar{\nu}$ as the function $\bar{\nu}(t, q, y, u) = g_0(t) + g_1(t)q + g_2(t)y + g_3(t)u$.

A.1. Proof of Theorem 3.2: Major Agent's Value Function. Here, we follow similar arguments to the ones in [21]. By employing a standard dynamic programming argument and the use of Itô's formula we obtain the following backward in time Hamilton–Jacobi–Bellman (HJB) equation

$$(A.1) \quad \begin{cases} \partial_t H_0(t, x, f, q, y, u) + \sup_{\nu_0 \in \mathcal{A}} \mathcal{L}^{\nu_0} H_0(t, x, f, q, y, u) - \phi q^2 = 0, \\ H_0(T, x, f, q, y, u) = x + fq - \alpha q^2, \end{cases}$$

where $\mathcal{L}^{\nu_0}$ is the infinitesimal generator of $(\bar{X}_t, \bar{F}_t, Q_t^{\nu_0})$ satisfying (3.4), and acts as follows

$$\begin{aligned} \mathcal{L}^{\nu_0} H_0 = & -(f + a_0 \nu_0) \nu_0 \partial_x H_0 + \nu_0 \partial_q H_0 \\ & + (\lambda_0 \nu_0 + \lambda \bar{\nu} + \lambda_\mu \mu_t) \partial_f H_0 + \bar{\nu} \partial_y H_0 + \mathcal{L}^\mu H_0 + \zeta H_0, \end{aligned}$$

where ζ is the operator that encompasses the higher moments of the fundamental price dynamics f , and $\mathcal{L}^\mu$ is the infinitesimal generator for the noise traders' process μ_t in the HJB equation. The optimal control in feedback form is

$$(A.2) \quad \nu_0^* := \arg \max_{\nu_0 \in \mathcal{A}} \mathcal{L}^{\nu_0} H_0 = \frac{(\lambda_0 \partial_f + \partial_q - f \partial_x) H_0}{2 a_0 \partial_x H_0}.$$

which gives

$$\begin{aligned} \mathcal{L}^{\nu_0^*} H_0 = & \frac{[(\lambda_0 \partial_f + \partial_q - f \partial_x) H_0]^2}{4 a_0 \partial_x H_0} \\ & + (\lambda(g_0 + g_1 q + g_2 y + g_3 u) + \lambda_\mu \mu_t) \partial_f H_0 + \bar{\nu} \partial_y H_0 + \mathcal{L}^\mu H_0 + \zeta H_0. \end{aligned}$$

Substitution into the HJB equation (A.1) yields

$$(A.3) \quad \begin{aligned} & (\partial_t + \lambda \bar{\nu} + \lambda_\mu u + \partial_f + \bar{\nu} \partial_y + \mathcal{L}^\mu) H_0(t, x, f, q, y, u) \\ & + \frac{[(\lambda_0 \partial_f + \partial_q - f \partial_x) H_0(t, x, f, q, y, u)]^2}{4 a_0 \partial_x H_0(t, x, f, q)} - \phi q^2 = 0, \end{aligned}$$

subject to the terminal condition $H_0(T, x, f, q, y, u) = x + fq - \alpha q^2$. The terminal condition, and form of this non-linear PDE suggests the ansatz of the form $H_0(t, x, f, q) = x + fq + h(t, q, y, u)$ and we find that h solves

$$(A.4) \quad (\partial_t + \bar{\nu} \partial_y + \mathcal{L}^\mu) h(t, q, y, u) + \lambda(\bar{\nu} + \lambda_\mu u) q + \frac{(\lambda_0 q + \partial_q h(t, q, y, u))^2}{4 a_0} - \phi q^2 = 0,$$

with terminal condition $h(T, q, y, u) = -\alpha q^2$. As q appears at most quadratic, we next apply the ansatz $h(t, q, y, u) = h_0(t, y, u) + h_1(t, y, u) q + h_2(t) q^2$. Substituting in (A.4) and collecting powers of q yields the coupled system of ODEs following equations (3.8a)-(3.8c).

Substituting this ansatz into (A.2) yields (3.9) for the optimal speed of trading.

Corollary A.1. Assume that the value function $h_1(t, y, u)$ for the stochastic optimal control problem (3.9) is a C^1 function of the variables t , y and u . Then, the major agent's auxiliary value functions (3.8b) and (3.8a) admit the ansatz

$$\begin{aligned} h_0(t, y, u) &= h_0^0(t) + h_0^1(t) y + h_0^2(t) y^2 + h_0^\mu(t) u y + h_0^{0,\mu} u + h_0^{2,\mu} u^2 \\ h_1(t, y, u) &= h_1^0(t) + h_1^1(t) y + h_1^\mu(t) u \end{aligned}$$

and the system of PDEs above reduce to the following system of ODEs

$$\begin{aligned}
(\text{A.5a}) \quad & \partial_t h_0^0 + g_0 h_0^1 + \frac{1}{4a_0} (h_1^0)^2 = 0, & h_0^0(T) = 0, \\
(\text{A.5b}) \quad & \partial_t h_0^1 + 2g_0 h_0^2 + g_2 h_0^1 + \frac{1}{2a_0} h_1^0 h_1^1 = 0, & h_0^1(T) = 0, \\
(\text{A.5c}) \quad & \partial_t h_0^2 + 2g_2 h_0^2 + \frac{1}{4a_0} (h_1^1)^2 = 0, & h_0^2(T) = 0, \\
(\text{A.5d}) \quad & (\partial_t - \kappa) h_0^\mu + 2g_3 h_0^2 + \frac{1}{2a_0} h_1^\mu h_1^1 = 0, & h_0^\mu(T) = 0, \\
(\text{A.5e}) \quad & (\partial_t + \lambda g_3 h_0^\mu - \kappa) h_0^{0,\mu} + \frac{1}{2a_0} h_1^0 h_1^\mu = 0, & h_0^{0,\mu}(T) = 0, \\
(\text{A.5f}) \quad & (\partial_t - \kappa) h_0^{2,\mu} + \frac{1}{2a_0} h_1^1 h_1^\mu = 0, & h_0^{2,\mu}(T) = 0, \\
(\text{A.5g}) \quad & \partial_t h_1^0 + g_0 h_1^1 + h_0^1 g_1 + \lambda g_0 + \frac{1}{2a_0} (\lambda_0 + 2h_2) h_1^0 = 0, & h_1^0(T) = 0, \\
(\text{A.5h}) \quad & \partial_t h_1^1 + 2h_0^2 g_1 + \lambda g_2 + \frac{1}{2a_0} (\lambda_0 + 2h_2 + 2a_0 g_2) h_1^1 = 0, & h_1^1(T) = 0, \\
(\text{A.5i}) \quad & (\partial_t - \kappa) h_1^\mu + g_3 h_1^1 + (\lambda g_3 + \lambda_\mu) + \frac{1}{2a_0} (\lambda_0 + 2h_2) h_1^\mu = 0, & h_1^\mu(T) = 0.
\end{aligned}$$

A.2. Proof of Theorem 3.2: Minor Agents' Value Function. Here, we follow similar arguments to the ones in [21]. By employing a standard dynamic programming argument and the use of Itô's formula we obtain the following backward in time HJB equation

$$(\text{A.6}) \quad \begin{cases} \partial_t H_i(t, x, f, q_i, q, y, u) + \sup_{\nu_i \in \mathcal{A}} \mathcal{L}^{\nu_i} H_i(t, x, f, q_i, q, y, u) = 0, \\ H_i(T, x, f, q_i, q) = x + f q_i - \psi q_i^2, \end{cases}$$

where $\mathcal{L}^{\nu_i}$ is the infinitesimal generator of $(\bar{X}_t^{\nu_i}, \bar{F}_t, Q_t^{\nu_i^*}, \bar{Q}_t^{\nu_i}, \mu_t)$ satisfying (3.5), and acts as follows

$$\mathcal{L}^{\nu_i} H_i = [-(f + a \nu_i) \nu_i \partial_x + \nu_i \partial_{q_i} + (\lambda_0 \nu_0^* + \lambda \bar{\nu} + \lambda_\mu \mu_t) \partial_f + \nu_0 \partial_q + \bar{\nu} \partial_y + \mathcal{L}^\mu H_i + \zeta] H_i.$$

The optimal control in feedback form is

$$(\text{A.7}) \quad \nu_i^* := \arg \max_{\nu_i \in \mathcal{A}} \mathcal{L}^{\nu_i} H_i = \frac{(\partial_q - f \partial_x) H_i}{2a \partial_x H_i},$$

which gives

$$\mathcal{L}^{\nu_i^*} H_i = \frac{[(\partial_q - f \partial_x) H_i]^2}{4a \partial_x H_i} + (\lambda_0 \nu_0^* + \lambda \bar{\nu} + \lambda_\mu \mu_t) \partial_f H_i + \mathcal{L}^\mu H_i + \nu_0 \partial_q H_i + \bar{\nu} \partial_y H_i + \zeta H_i.$$

Substituting into the HJB equation (A.6) yields

$$\partial_t H_i + \frac{[(\partial_q - f \partial_x) H_i]^2}{4a \partial_x H_i} + (\lambda_0 \nu_0^* + \lambda \bar{\nu} + \lambda_\mu \mu_t) \partial_f H_i + \mathcal{L}^\mu H_i + \nu_0 \partial_q H_i + \bar{\nu} \partial_y H_i + \zeta H_i = 0,$$

with terminal condition $H_i(T, x, f, q_i, q, y, u) = x + f q_i - \psi q_i^2$. We next separate the book value of the cash and assets in hand from value of optimally trading the remaining shares by applying the ansatz $H_i(t, x, f, q_i, q) = x + f q_i + l(t, q_i, q, y, u)$ where h solves the PDE

$$(\text{A.8}) \quad \partial_t l + \frac{1}{4a} (\partial_{q_i} l)^2 + q (\lambda_0 \nu_0^* + \lambda \bar{\nu} + \lambda_\mu \mu_t) + \mathcal{L}^\mu l + \nu_0 \partial_q l + \bar{\nu} \partial_y l = 0,$$

with terminal condition $l(T, q_i, q, y, u) = -\psi q_i^2$. Next, noticing that the terminal condition and the PDE is at most quadratic in q , we apply a further ansatz $l(t, q_i, q, y, u) = l_0(t, q, y, u) + l_1(t, q, y, u) q_i + l_2(t) q_i^2$

to (A.8) yields the following coupled system of PDEs

$$(A.9a) \quad (\partial_t + \nu_0 \partial_q + \bar{\nu} \partial_y + \mathcal{L}^\mu) l_0(t) + \frac{1}{4a} l_1^2(t) = 0, \quad l_0(T, y, u) = 0$$

$$(A.9b) \quad (\partial_t + \nu_0 \partial_q + \bar{\nu} \partial_y + \mathcal{L}^\mu) l_1(t) + \frac{1}{a} (l_1(t) l_2(t)) + (\lambda_0 \nu_0 + \lambda \bar{\nu} + \lambda_\mu \mu) = 0, \quad l_1(T, y, u) = 0$$

$$(A.9c) \quad \partial_t l_2(t) + \frac{1}{a} (l_2(t))^2 = 0, \quad l_2(T) = -\psi.$$

where we denote

$$\nu_0 = \nu_0(t, q, y, u) \quad \text{and} \quad \bar{\nu} = \bar{\nu}(t, q, y, u).$$

Finally, substituting the sequence of ansatz into the feedback form for the optimal control in (A.7) leads directly to (3.11).

Corollary A.2. *Assume that the value function $H_i(t, x, f, q)$ for the auxiliary stochastic optimal control problem (3.5) is a C^1 function of the variables t , x and q , and C^2 function of variable f . Then, a generic minor agent's auxiliary value function (A.9b) admits the ansatz*

$$l_1(t, q, y, u) = l_1^0(t) + l_1^1(t) q + l_1^2(t) y + l_1^\mu(t)$$

which results in the set of PDEs

$$(A.10a) \quad \partial_t l_1^0(t) + g_0(t) l_1^2(t) + \frac{h_1^0(t)}{2a_0} (\lambda_0 + l_1^0(t)) + \frac{1}{a} (l_1^0(t) l_2(t)) + \lambda g_0(t) = 0, \quad l_1^0(T) = 0,$$

$$(A.10b) \quad \partial_t l_1^1(t) + \frac{1}{a} (l_1^1(t) l_2(t)) + g_1(t) (\lambda + l_1^2(t)) + \frac{h_1^1(t)}{2a_0} (l_1^1(t) + \lambda_0) = 0, \quad l_1^1(T) = 0,$$

$$(A.10c) \quad \partial_t l_1^2(t) + \frac{1}{a} (l_1^2(t) l_2(t)) + g_2(t) l_1^2(t) + \frac{\lambda_0 + 2h_2}{2a_0} (\lambda_0 + l_1^2(t)) + (\lambda g_2(t)) = 0, \quad l_1^2(T) = 0,$$

$$(A.10d) \quad (\partial_t - \kappa) (l_1^\mu(t) + g_3(t) l_1^2(t) + \frac{h_1^\mu(t)}{2a_0} l_1^\mu(t) + \frac{1}{a} l_1^\mu(t) l_2(t) + \lambda g_3(t) + \lambda_\mu) = 0, \quad l_1^\mu(T) = 0.$$

Appendix B. Proof of ϵ -Nash Equilibrium.

In this appendix, we interpret $\bar{\nu}$ as the function $\bar{\nu}(t, q, y, u) = g_0(t) + g_1(t)q + g_2(t)y + g_3(t)u$.

B.1. Proof of Lemma 5.1: Bounds on Fundamental Price and Cash Processes. We prove the minor agent inequalities (5.15a) and (5.15b) first, then the major agent inequalities (5.15c) and (5.15d).

Firstly, there is a constant C such that

$$(B.1) \quad \int_0^T ((\nu_{i,t})^2 + (\nu_{i,t}^*)^2) dt < C,$$

as $J(\nu_i)$ is quadratic concave in ν_i , so for large enough C , we have that $\int_0^T (\nu_{i,t})^2 dt > C \implies J(\nu_i) < J(\nu_i^*)$.

Next, the triangle inequality implies that the difference between the finite population fundamental price with an arbitrary control for the i -th agent, F_{-i}^* , and the mean-field fundamental price, $\bar{F}^*$, satisfies

$$(B.2) \quad \mathbb{E} |F_{-i,t}^* - \bar{F}_t^*| \leq \mathbb{E} |F_{-i,t}^* - F_t^*| + \mathbb{E} |F_t^* - \bar{F}_t^*|,$$

where F^* is given in (5.3). From (5.3), (5.6) and (5.12) we respectively have

$$\begin{aligned} F_t^* &= F_0 + \int_0^t \left(\lambda_0 \nu_{0,t} + \frac{\lambda}{N} \sum_{i \in \mathfrak{N}} \nu_{i,t}^* \right) dt + M_t, \\ \bar{F}_t^* &= F_0 + \int_0^t \left(\lambda_0 \nu_{0,t}^* + \lambda(g_0(t) + g_1(t) Q_{0,t} + g_2(t) \mathcal{Q}_t) \right) dt + M_t, \quad \text{and} \\ F_{-i,t}^* &= F_0 + \int_0^t \left(\lambda_0 \nu_{0,t}^* + \frac{\lambda}{N} \sum_{j \in \mathfrak{N}, j \neq i} \nu_{j,t}^* + \frac{\lambda}{N} \nu_{i,t} \right) dt + M_t. \end{aligned}$$

Using this representation, and by combining the inequality (B.1), with Cauchy-Schwartz and Jensen's inequality, we have

$$\begin{aligned} \mathbb{E} |F_{-i,t}^* - F_t^*| &= \mathbb{E} \left| \frac{\lambda}{N} \int_0^t (\nu_{i,t} - \nu_{i,t}^*) dt \right| \\ (B.3) \quad &\leq \frac{\lambda}{N} \mathbb{E} \left| \sqrt{\int_0^t ((\nu_{i,t})^2 + (\nu_{i,t}^*)^2) dt} \right| < C \frac{\lambda}{N} = O(N^{-1}). \end{aligned}$$

The equality

$$(B.4) \quad \mathbb{E} |F_t^* - \bar{F}_t^*| = \mathbb{E} \left| \lambda \int_0^t \left(N^{-1} \sum_{i=1}^N \nu_{i,t}^* - \bar{\nu}_t^* \right) \right|$$

also holds. From the definition of the trading speed, we have

$$\begin{aligned} &\mathbb{E} \left| \int_0^t \left(N^{-1} \sum_{i=1}^N \nu_{i,s}^* - \bar{\nu}_s^* \right) ds \right| \\ &\leq \mathbb{E} \left| \int_0^t \left(- \left(\frac{a}{\psi} + (T-s) \right)^{-1} + \frac{1}{2a} l_1^2(s) \right) \left(N^{-1} \sum_{i=1}^N Q_{i,s} - \mathcal{Q}_s \right) ds \right| \\ &\leq C \left| N^{-1} \sum_{i=1}^N Q_{i,0} - \mathcal{Q}_0 \right| \leq \lambda C T \epsilon_N \end{aligned}$$

by Assumption (A3), and Gronwall's inequality, and where ϵ_N is defined in (5.4).

Therefore, we obtain the inequality

$$(B.5) \quad \mathbb{E} |F_t^* - \bar{F}_t^*| = \lambda \mathbb{E} \left| \int_0^t \left(N^{-1} \sum_{i=1}^N \nu_{i,t}^* - \bar{\nu}_t^* \right) \right| \leq \lambda C T \epsilon_N.$$

Throughout, to simplify notation, the constant C may differ in the various inequalities. Hence, this together with (B.2) and (B.3) implies that

$$(B.6) \quad \mathbb{E} |F_{-i,t}^* - \bar{F}_t^*| = O(\epsilon_N) + O(N^{-1}), \quad t \in \mathfrak{T},$$

and proves (5.15a).

Next, from (5.5) and (5.14) we have

$$\bar{X}_{-i,t}^* = X_{i,0} - \int_0^t \left(\bar{F}_t^* + a \nu_{i,t} \right) \nu_{i,t} dt \quad \text{and} \quad X_{-i,t}^* = X_{i,0} - \int_0^t \left(F_{-i,t}^* + a \nu_{i,t} \right) \nu_{i,t} dt.$$

As $\int_0^T (\nu_{i,t})^2 dt < C$, from Cauchy-Schwartz and (B.6), we have

$$\mathbb{E} \left| X_{-i,t}^* - \bar{X}_{-i,t}^* \right| = \mathbb{E} \left| \int_0^t (\bar{F}_t^* - F_{-i,t}^*) \nu_{i,t} dt \right| \leq CT (O(\epsilon_N) + O(N^{-1})),$$

and proves (5.15b).

For the major agent, from (5.9) and (5.11), respectively, we have

$$\begin{aligned} \bar{F}_{-0,t}^* &= F_0 + \int_0^t (\lambda_0 \nu_{0,t} + \lambda \bar{\nu}_t^*) dt + M_t \quad \text{and} \\ F_{-0,t}^* &= F_0 + \int_0^t \left(\lambda_0 \nu_{0,t} + \frac{\lambda}{N} \sum_{i=1}^N \nu_{i,t}^* \right) dt + M_t. \end{aligned}$$

Following arguments that parallel those used in proving (B.5), we obtain

$$(B.7) \quad \mathbb{E} |F_{-0,t}^* - \bar{F}_{-0,t}^*| \leq C' \epsilon_N,$$

for some other constant C' . Next, from (5.10) and (5.13) we have

$$\begin{aligned} \bar{X}_{-0,t}^* &= X_{0,0} - \int_0^t (\bar{F}_{-0,t}^* + a_0 \nu_{0,t}) \nu_0(t, q, y) dt \quad \text{and} \\ X_{-0,t}^* &= X_{0,0} - \int_0^t (F_{-0,t}^* + a_0 \nu_{0,t}) \nu_0(t, q, y) dt. \end{aligned}$$

Therefore,

$$\mathbb{E} |X_{-0,t}^* - \bar{X}_{-0,t}^*| = \mathbb{E} \left| \int_0^t (\bar{F}_{-0,t}^* - F_{-0,t}^*) \nu_{0,t} dt \right| \leq C'' \epsilon_N$$

using (B.7), (B.1) and Cauchy-Schwartz. This together with (B.7) concludes the proof.

B.2. Proof of Theorem 5.1: The MFG strategy is a ϵ -Nash Equilibrium. For this proof, the fact that ν_0, ν_i are functions of q_i, q, y are purposely omitted for clarity.

The left inequality is trivial. To establish the right inequality, we consider following two cases separately.

Case I. Let us assume that a representative minor agent $i \in \mathfrak{N}$ uses an alternative strategy $\nu_i \in \mathcal{A}$ while all other agents apply their MFG control laws. Then, the performance criteria of the minor agent defined in (2.10) is

$$(B.8) \quad J_i(\nu_i; \nu_{-i}^*) = \mathbb{E} [X_{-i,T}^* + Q_{i,T}(F_{-i,T}^* - \psi Q_{i,T})]$$

where the dynamics of $F_{-i}^*(\cdot)$ and $X_{-i}^*(\cdot)$ are given in (5.12) and (5.14), respectively. We make the decomposition

$$(B.9) \quad J_i(\nu_i; \nu_{-i}^*) = \mathbb{E} [\bar{X}_{-i,T}^* + Q_{i,T}(\bar{F}_T^* - \psi Q_{i,T}) + (X_{-i,T}^* - \bar{X}_{-i,T}^*) + Q_{i,T}(F_{-i,T}^* - \bar{F}_T^*)]$$

where $\bar{X}_{-i}^*(\cdot)$ and $\bar{F}^*(\cdot)$ are given in (5.5) and (5.6), respectively. The linearity of the expectation operator implies that

$$\begin{aligned} J_i(\nu_i; \nu_{-i}^*) &= \mathbb{E} [\bar{X}_{-i,T}^* + Q_{i,T}(\bar{F}_T^* - \psi Q_{i,T})] + \mathbb{E} [X_{-i,T}^* - \bar{X}_{-i,T}^*] + \mathbb{E} [Q_{i,T}(F_{-i,T}^* - \bar{F}_T^*)] \\ (B.10) \quad &:= \bar{J}_i(\nu_i) + I_1 + I_2 \end{aligned}$$

where $\bar{J}_i(\nu_i)$ is the utility function of the auxiliary stochastic optimal control problem of the generic minor agent defined in (3.5). But, $\nu_i^*(\cdot) \in \mathcal{A}$ is the optimal control with respect to the utility function $\bar{J}_i(\nu_i)$, i.e. $\nu_i^* = \arg \sup_{\nu_i \in \mathcal{A}} \bar{J}_i(\nu_i)$, by the construction of the MFG system (4.1) in Section 4.1. Therefore, we have

$$(B.11) \quad \bar{J}_i(\nu_i) \leq \bar{J}_i(\nu_i^*),$$

$$(B.12) \quad I_1 = O(\epsilon_N) + O(N^{-1}),$$

where (B.12) follows from Lemma 5.1. Moreover, the Cauchy-Schwarz inequality gives

$$(B.13) \quad |I_2| \equiv \left| \mathbb{E}[Q_{i,t}(F_{-i,T}^* - \bar{F}_T^*)] \right| \leq \left(\mathbb{E}|Q_{i,t}|^2 \right)^{1/2} \times \left(\mathbb{E}|F_{-i,T}^* - \bar{F}_T^*|^2 \right)^{1/2}.$$

Using the inequality $(x + y)^2 \leq 2x^2 + 2y^2$, and following similar arguments as in proving (B.3) and (B.6) in Lemma 5.1, we have

$$(B.14) \quad \left(\mathbb{E}|F_{-i,T}^* - \bar{F}_T^*|^2 \right)^{1/2} = O(\epsilon_N) + O(N^{-1}).$$

Additionally, by the Cauchy-Schwarz inequality, we have

$$\mathbb{E}|Q_{i,t}|^2 = \mathbb{E} \left| Q_{i,0} + \int_0^T \nu_{i,t} dt \right|^2 \leq 2\mathbb{E}|Q_{i,0}|^2 + 2T \int_0^T |v_i(t)|^2 dt \leq 2k + 2C < \infty,$$

where k is the second moment bound in assumption (A1). This together with (B.13) and (B.14) implies

$$(B.15) \quad I_2 = O(\epsilon_N) + O(N^{-1}).$$

From (B.10)-(B.12) and (B.15) we have

$$(B.16) \quad J_i(\nu_i; \nu_{-i}^*) = \bar{J}_i(\nu_i) + I_1 + I_2 \leq \bar{J}_i(\nu_i^*) + O(\epsilon_N) + O(N^{-1}).$$

Along the optimal strategy, we have

$$J_i(\nu_i^*; \nu_{-i}^*) = \mathbb{E} \left[X_{i,T}^* + Q_{i,T}^*(F_T^* - \psi Q_{i,T}^*) \right],$$

where $X_i^*(\cdot)$, $Q_i^*(\cdot)$ and $F^*(\cdot)$ are given in (5.2)-(5.3). As in (B.9) we make the decomposition

$$(B.17) \quad \begin{aligned} J_i(\nu_i^*; \nu_{-i}^*) &= \mathbb{E} \left[\bar{X}_{i,T}^* + Q_{i,T}^*(\bar{F}_T^* - \psi Q_{i,T}^*) + (X_{i,T}^* - \bar{X}_{i,T}^*) + Q_{i,T}^*(F_T^* - \bar{F}_T^*) \right] \\ &= \bar{J}_i(\nu_i^*) + \mathbb{E}[X_{i,T}^* - \bar{X}_{i,T}^*] + \mathbb{E}[Q_{i,T}^*(F_T^* - \bar{F}_T^*)], \end{aligned}$$

where $\bar{F}^*$ and $\bar{X}_i^*$ are defined in (5.6) and (5.8), respectively. Following the same argument in the proof of Lemma (B.6), and in (B.12) and (B.15) above we can show that

$$\left| \mathbb{E}[X_{i,T}^* - \bar{X}_{i,T}^*] + \mathbb{E}[Q_{i,T}^*(F_T^* - \bar{F}_T^*)] \right| = O(\epsilon_N).$$

This together with (B.17) gives

$$(B.18) \quad \left| J_i(\nu_i^*; \nu_{-i}^*) - \bar{J}_i(\nu_i^*) \right| = O(\epsilon_N).$$

Finally, (B.16) and (B.18) imply

$$J_i(\nu_i; \nu_{-i}^*) \leq J_i(\nu_i^*; \nu_{-i}^*) + O(\epsilon_N) + O(N^{-1}),$$

and concludes the right inequality (5.1) for the representative minor agent i .

Case II. Let us assume that the major agent uses an alternative strategy $\nu_0 \in \mathcal{A}$ while all minor agents apply the MFG trading laws. Then, we may write the wealth function of the major agent defined in (2.9) as

$$(B.19) \quad J_0(\nu_0; \nu_{-0}^*) = \mathbb{E} \left[X_{-0,T}^* + Q_{0,T} (F_{-0,T}^* - \psi Q_{0,T}) - \phi \int_0^T (Q_{0,s})^2 ds \right],$$

where F_{-0}^* and X_{-0}^* are given in (5.11) and (5.13), respectively. We make the decomposition

$$(B.20) \quad \begin{aligned} J_0(\nu_0; \nu_{-0}^*) &= \mathbb{E} \left[\bar{X}_{-0,T}^* + Q_{0,T} (\bar{F}_{-0,T}^* - \psi Q_{0,T}) - \phi \int_0^T (Q_{0,s})^2 ds \right. \\ &\quad \left. + (X_{-0,T}^* - \bar{X}_{-0,T}^*) + Q_{0,T} (F_{-0,t}^* - \bar{F}_{-0,t}^*) \right] \\ &= \bar{J}_0(\nu_0) + \mathbb{E} [X_{-0,T}^* - \bar{X}_{-0,T}^*] + \mathbb{E} [Q_{0,T} (F_{-0,t}^* - \bar{F}_{-0,t}^*)] := \bar{J}_0(\nu_0) + L_1 + L_2. \end{aligned}$$

where $\bar{F}_{-0}^*(\cdot)$ and $\bar{X}_{-0}^*(\cdot)$ are given in (5.9) and (5.10), respectively, and $\bar{J}_0(\nu_0)$ is the utility function of the auxiliary stochastic optimal control problem of the major agent defined in (3.4). But, $\nu_0^*(\cdot) \in \mathcal{A}$ is the optimal control with respect to the utility function $\bar{J}_0(\nu_0)$, i.e. $\nu_0^* = \arg \sup_{\nu_0 \in \mathcal{A}} \bar{J}_0(\nu_0)$, by the construction of the MFG system (4.1) in Section 4.1. We note that the postulated mean-field form (3.3) let us get the result that $\nu_0^* = \arg \sup_{\nu_0 \in \mathcal{A}} \bar{J}_0(\nu_0)$ above, regardless the difference between $\bar{F}(\cdot)$ given in (3.4) and $\bar{F}_{-0}^*(\cdot)$ in (5.9).

Following arguments exactly parallel to those used in *Case I* above yield

$$(B.21) \quad L_1 + L_2 = O(\epsilon_N), \quad \text{and} \quad \left| J_0(\nu_0^*; \nu_{-0}^*) - \bar{J}_0(\nu_0^*) \right| = O(\epsilon_N).$$

Finally, (B.20)-(B.21) imply that

$$J_0(\nu_0; \nu_{-0}^*) \leq J_0(\nu_0^*; \nu_{-0}^*) + O(\epsilon_N)$$

which concludes the right inequality (5.1) for the major agent. This completes the proof.

Appendix C. Simulation parameters. In our simulations, we assigned the following parameters based on our estimated parameters $\lambda_0 = \lambda = \hat{b}$, $a_0 = 10$, $a = \hat{k}$. We use parameters estimated by [20] (Tables 7 and 8) and record the relevant parameters for our simulations in Table 4 for convenience.

Table 4
Simulation Parameters extracted from Tables 7 and 8 of [20].

Ticker	Mid-Price	σ	$\hat{b} (\times 10^{-6})$	$\hat{k} (\times 10^{-6})$	λ_{MO}^+	λ_{MO}^-	$\mathbb{E}[\eta^+]$	$\mathbb{E}[\eta^-]$
EBAY	53.41	0.17	4.49	2.50	549.41	585.13	358.36	347.18
INTC	30.26	0.16	1.09	0.44	438.86	453.91	1,048.99	1,013.83
DOW	49.48	0.23	6.87	3.01	389.76	400.67	282.28	286.41
CSCO	24.21	0.14	0.63	0.24	306.70	307.99	1,500.35	1,376.49
COST	121.37	0.12	25.30	10.60	294.98	293.71	139.89	141.38
AXP	89.78	0.19	17.20	8.95	281.10	273.04	159.27	158.29
AMAT	20.85	0.23	1.60	0.60	248.86	247.66	861.38	833.01
PNC	84.67	0.15	19.90	10.00	206.79	209.22	136.35	143.73
AA	14.44	0.27	1.49	0.59	154.65	152.89	953.22	990.34
FMER	19.17	0.21	8.74	3.14	104.04	105.58	254.01	247.77
CNQR	105.96	0.29	88.60	76.20	96.47	99.96	128.79	136.23
ROC	76.10	0.19	43.40	56.10	96.00	93.21	114.76	117.90
IMGN	12.19	0.50	15.90	9.97	90.61	87.21	218.09	221.55
DK	30.83	0.78	45.50	48.30	80.26	86.34	118.25	117.12
ARCC	16.96	0.12	2.83	1.32	54.83	55.46	583.79	545.10
BXS	22.88	0.26	27.20	17.20	47.64	48.19	143.36	143.23
AYI	128.21	0.31	203.00	656.00	41.57	40.66	106.51	107.41

Appendix D. Almgren-Chriss Framework.

In this paper, the Almgren-Chriss (AC) framework refers to an agent that aims to completely liquidate her inventories $Q_0^{\nu_0} = (Q_{0,t}^{\nu_0})_{t \in \mathfrak{T}}$ by controlling her continuous rate of trading $\nu_0 = (\nu_{0,t})_{t \in \mathfrak{T}}$ so that $dQ_{0,t}^{\nu_0} = \nu_{0,t} dt$, $t \in \mathfrak{T}$. The fundamental price $F^{\nu_0} = (F_t^{\nu_0})_{t \in \mathfrak{T}}$ (i.e. midprice of the asset) at time t is $F_t^{\nu_0} = \lambda_0 (Q_{0,t}^{\nu_0} - Q_{0,0}^{\nu_0}) + \sigma W_t$ where $W = (W_t)_{t \in \mathfrak{T}}$ is $\mathbb{P}$ -Brownian motion. The agent's execution price from selling at rate $\nu_{0,t}$ is $S_{0,t}^{\nu_0} = F_t^{\nu_0} + a_0 \nu_{0,t}$, $t \in \mathfrak{T}$. The major agent's cash process $X_0^{\nu_0} = (X_{0,t}^{\nu_0})_{t \in \mathfrak{T}}$ satisfies the SDE $dX_{0,t}^{\nu_0} = -S_{0,t}^{\nu_0} \nu_{0,t} dt$, $t \in \mathfrak{T}$ with given $X_{0,0}^{\nu_0} = x_0$ – typically set to 0. The major agent's performance criterion is $J_0(\nu_0) := \mathbb{E} \left[X_{0,T}^{\nu_0} + Q_{0,T}^{\nu_0} (F_T^{\nu_0} - \alpha Q_{0,T}^{\nu_0}) - \phi \int_0^T (Q_{0,s}^{\nu_0})^2 ds \right]$ and we take $\alpha \rightarrow \infty$ to ensure complete liquidation. The optimal strategy is known to be $\nu_{0,t}^{*AC} = -\tanh(\gamma(T-t)) Q_{0,t}^{*AC}$, where $\gamma = \sqrt{\phi/\lambda_0}$ (see [3, 21]).

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